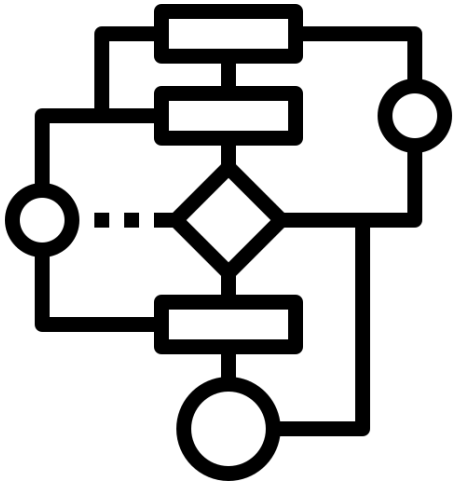
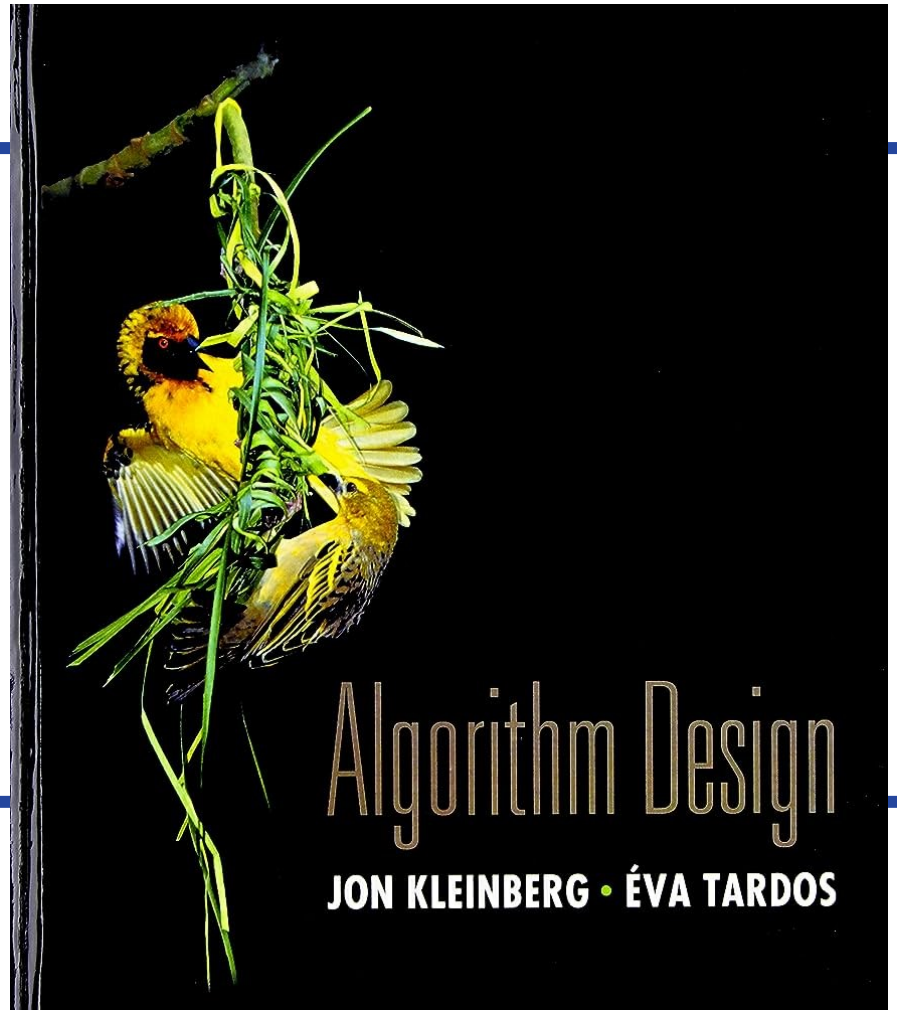


CS 310: Algorithms

Lecture 16

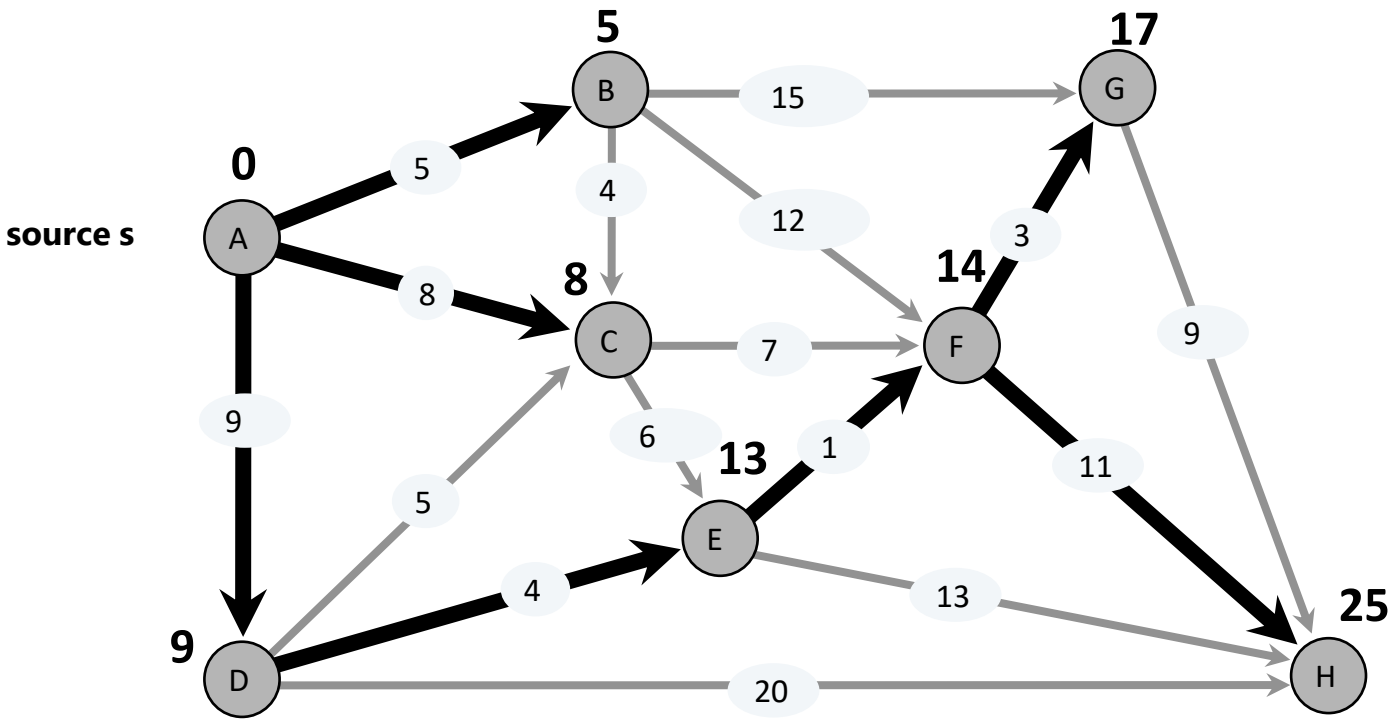
Instructor: Naveed Anwar Bhatti



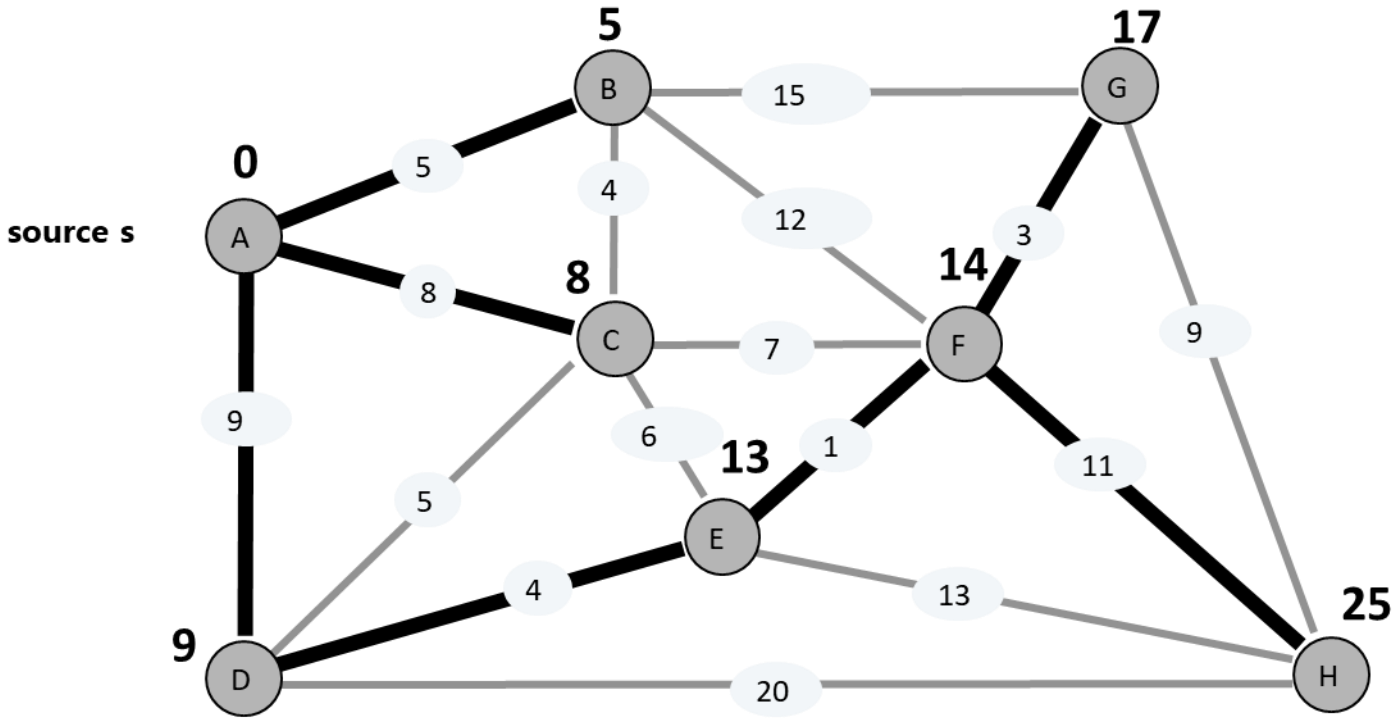
Chapter 4:
Greedy Algorithms

Section 4.5:
Minimums Spanning Tree (MST)

Single-source shortest paths problem (Dijkstra Algorithm)



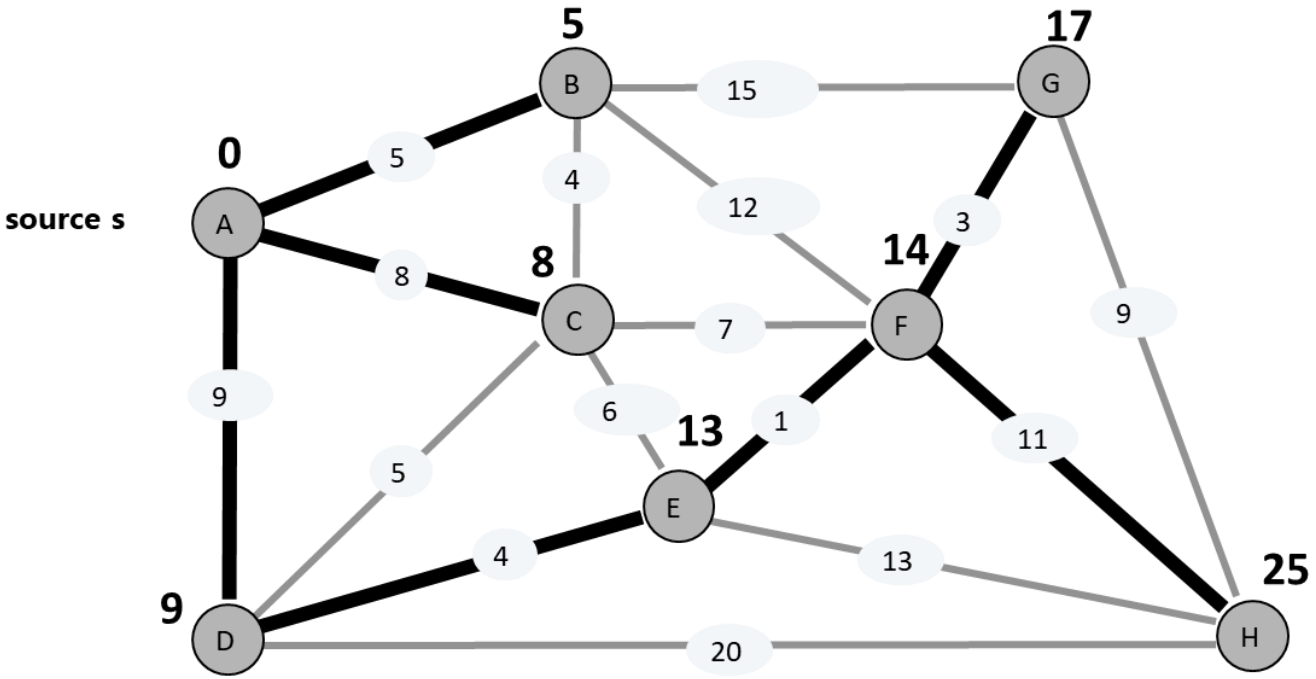
Single-source shortest paths problem (Dijkstra Algorithm)



Spanning tree definition

Def. Subgraph $H = (V, T)$ is a ***spanning tree*** of an undirected graph $G = (V, E)$ that is:

- **A tree:** *connected* and *acyclic*.
- **Spanning:** includes all of the vertices.



graph G = (V, E)

spanning tree H = (V, T)

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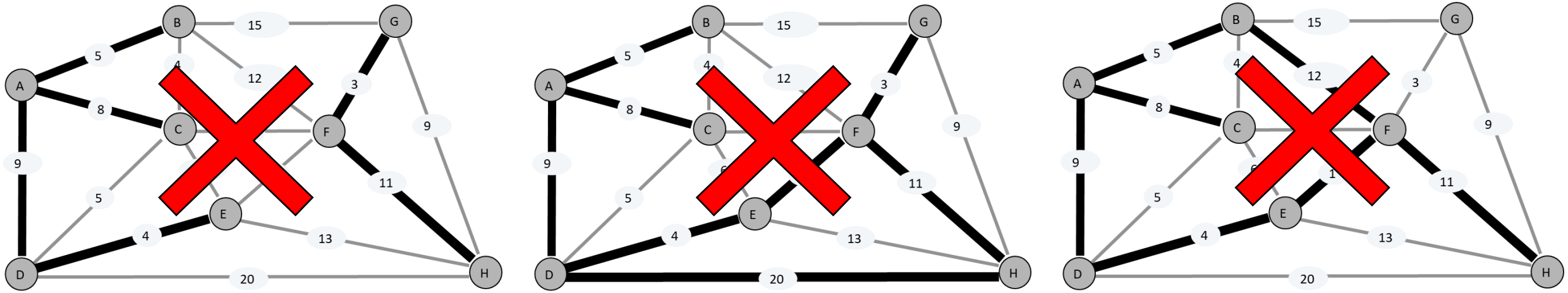


graph G = (V, E)

spanning tree H = (V, T)

Scan the QR or use link to join

https://forms.office.com/r/ZV0ZXt62Jy

Copy link

5 responses submitted

Let G be a connected edge-weighted graph with V vertices and E edges. How many edges will be in spanning tree of G?

V – 1	100%
V	0%
E – 1	0%
E	0%
I don't know.	0%

Treemap Bar

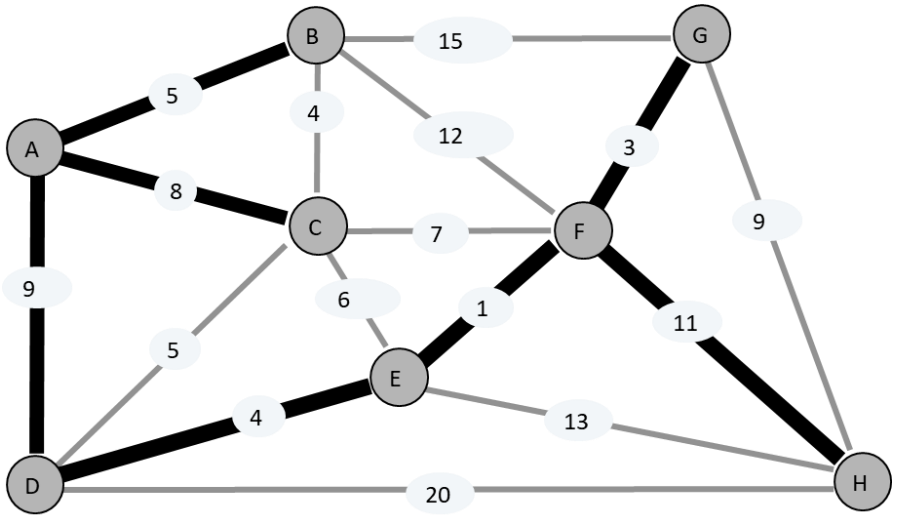
1 of 1

Spanning tree properties

- Proposition. Let $H = (V, T)$ be a **spanning tree** of an undirected graph $G = (V, E)$. Then, the following are true:
  - H is acyclic and connected.
  - H is connected and has $|V| - 1$ edges.
  - H is acyclic and has $|V| - 1$ edges.
  - H is minimally connected: removal of any edge disconnects it.
  - H is maximally acyclic: addition of any edge creates a cycle.

graph G = (V, E)

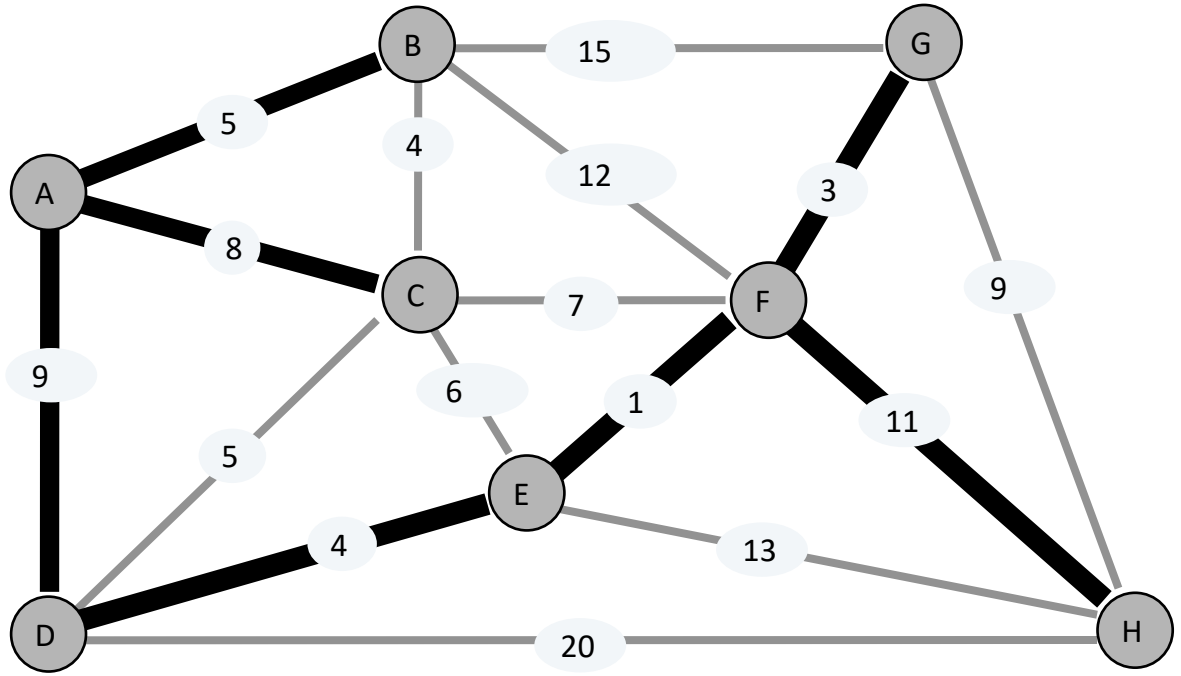
spanning tree H = (V, T)



Minimum Spanning Tree (MST)

- **Input.** Connected, undirected graph ***G*** with positive edge weights.
- **Output.** A spanning tree of minimum weight.

5+8+9+4+1+3+11 = 41



graph G = (V, E)

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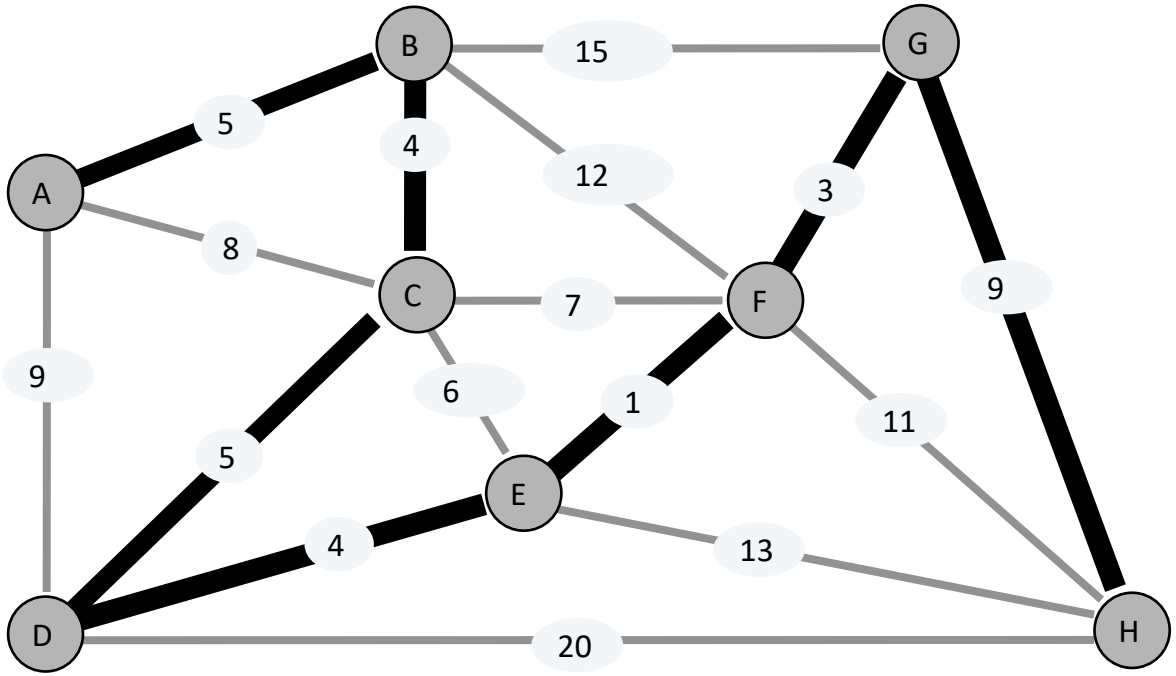
- **Output.** A spanning tree of minimum weight.

5+4+5+4+1+3+9 = 31

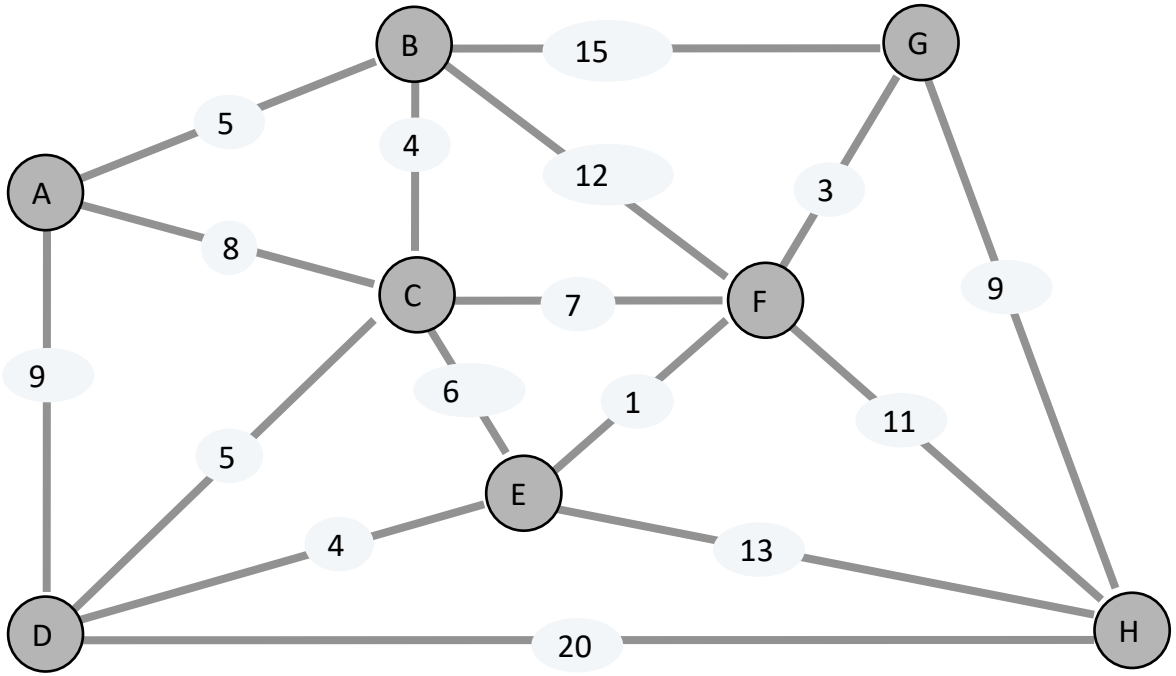


graph G = (V, E)

Minimum spanning tree H = (V, T)

Cut Property

Def: A **cut** in a graph is a partition of its vertices into two (nonempty) sets



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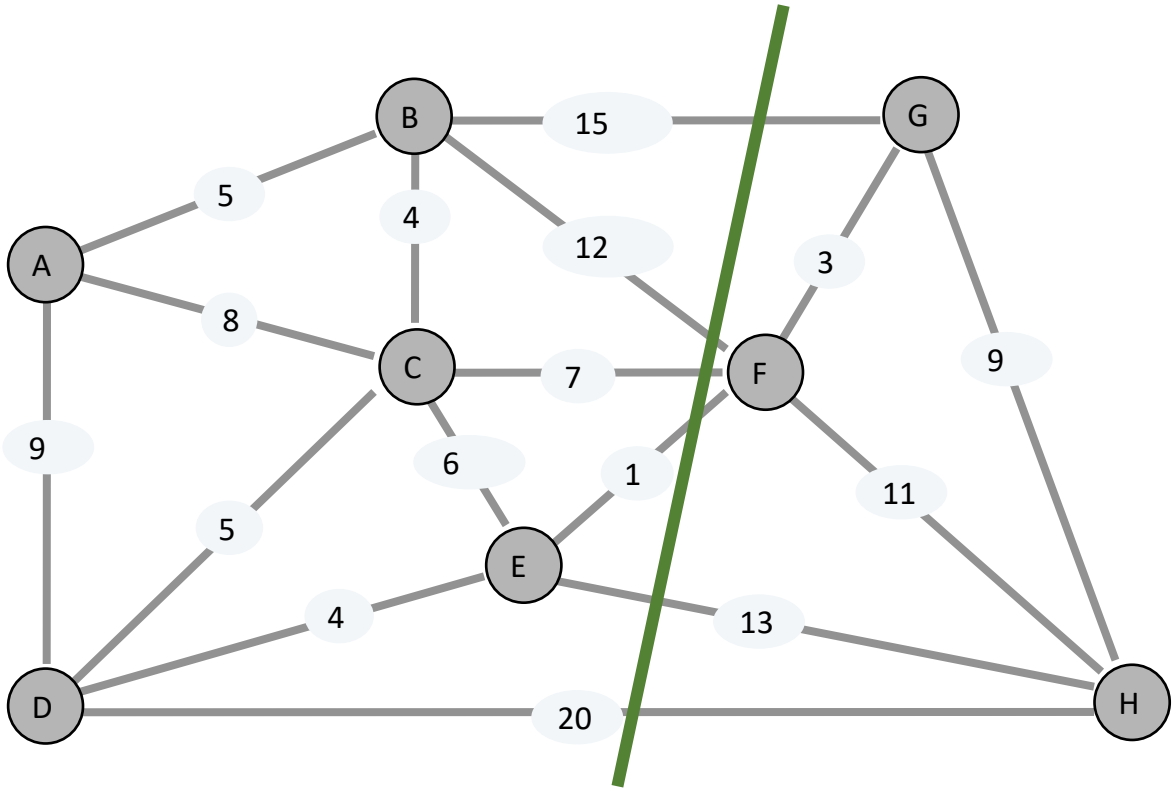

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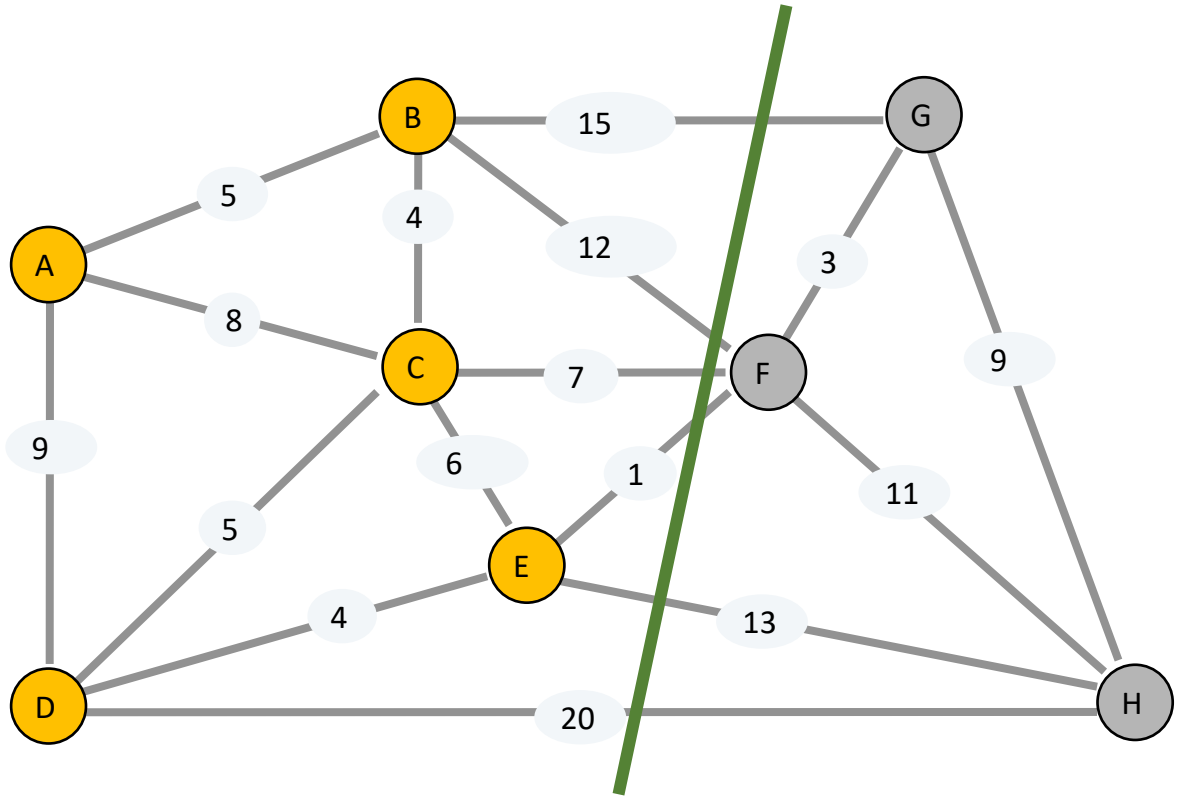
{A,B,C,D,E}



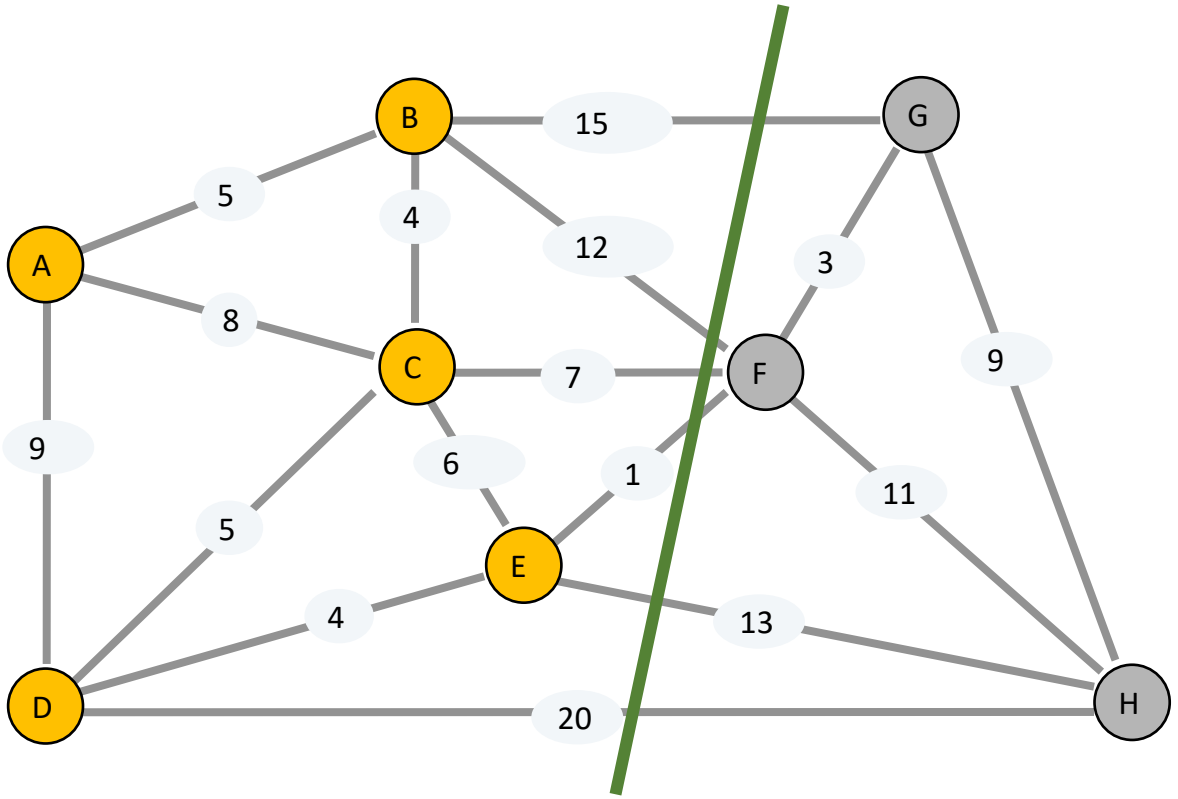
{F, G, H}

Cut Property

Def: A **cut** in a graph is a partition of its vertices into two (nonempty) sets

Def: A **crossing edge** connects a vertex in one set with a vertex in the other.

{A,B,C,D,E}



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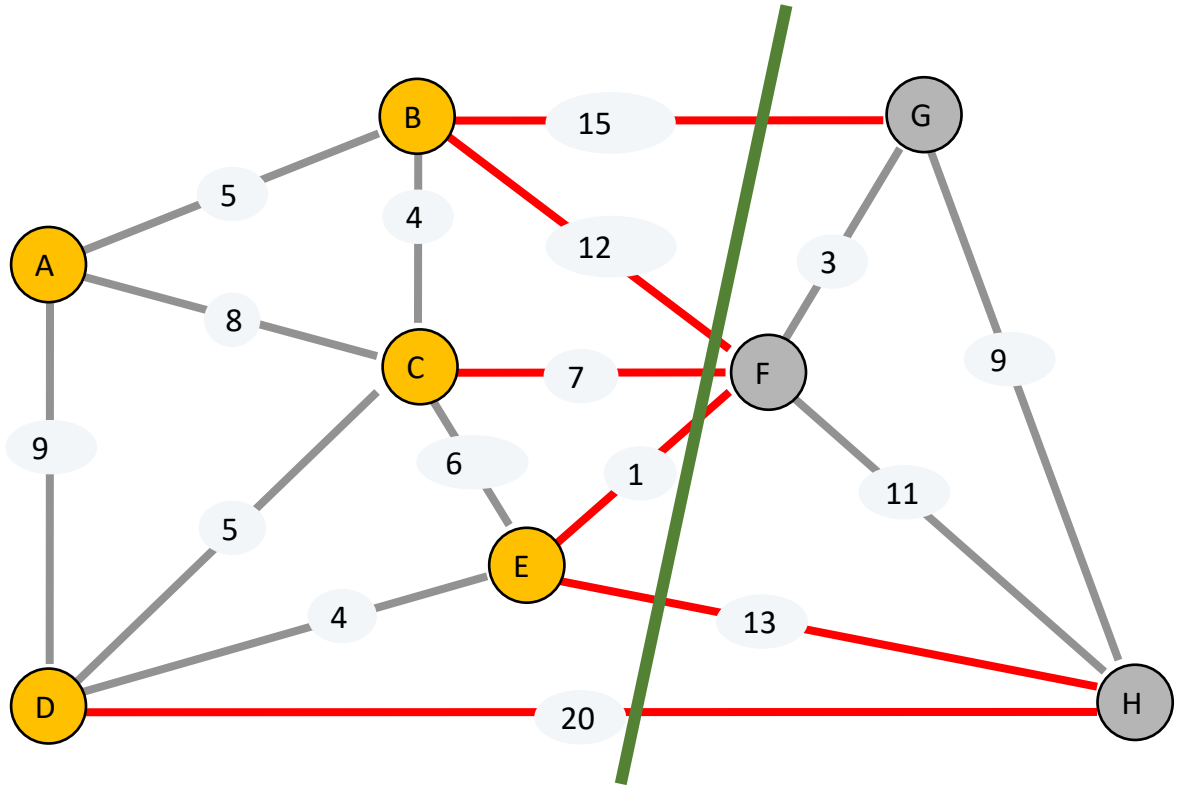

{F, G, H}

Cut property: Given any cut, the crossing edge of min weight is in the MST.

Cut Property

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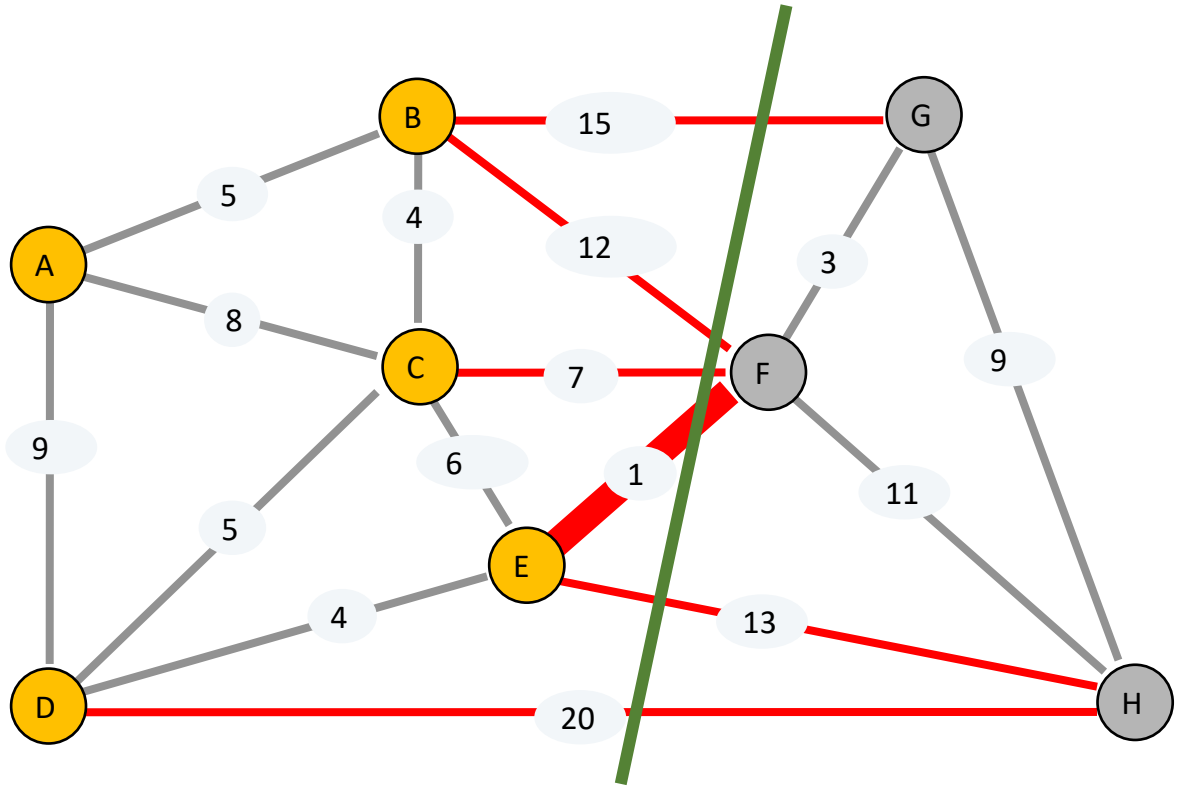
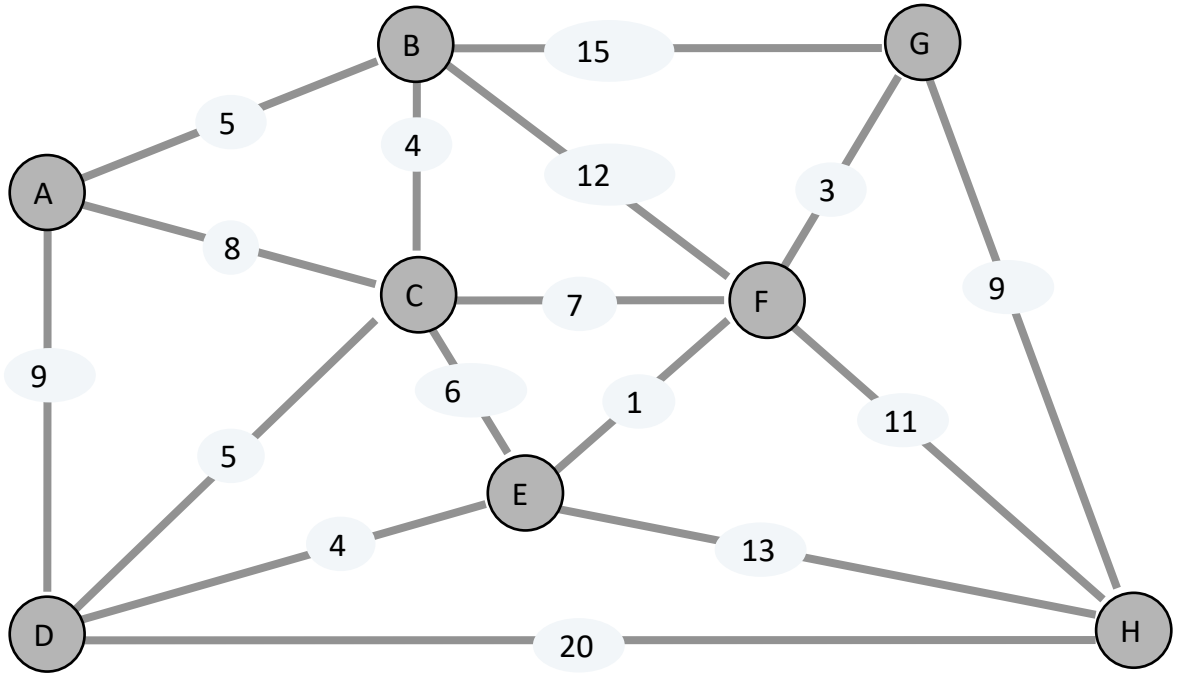


{F, G, H}

Cut property: Given any cut, the crossing edge of min weight is in the MST.

Prim's algorithm to find MST



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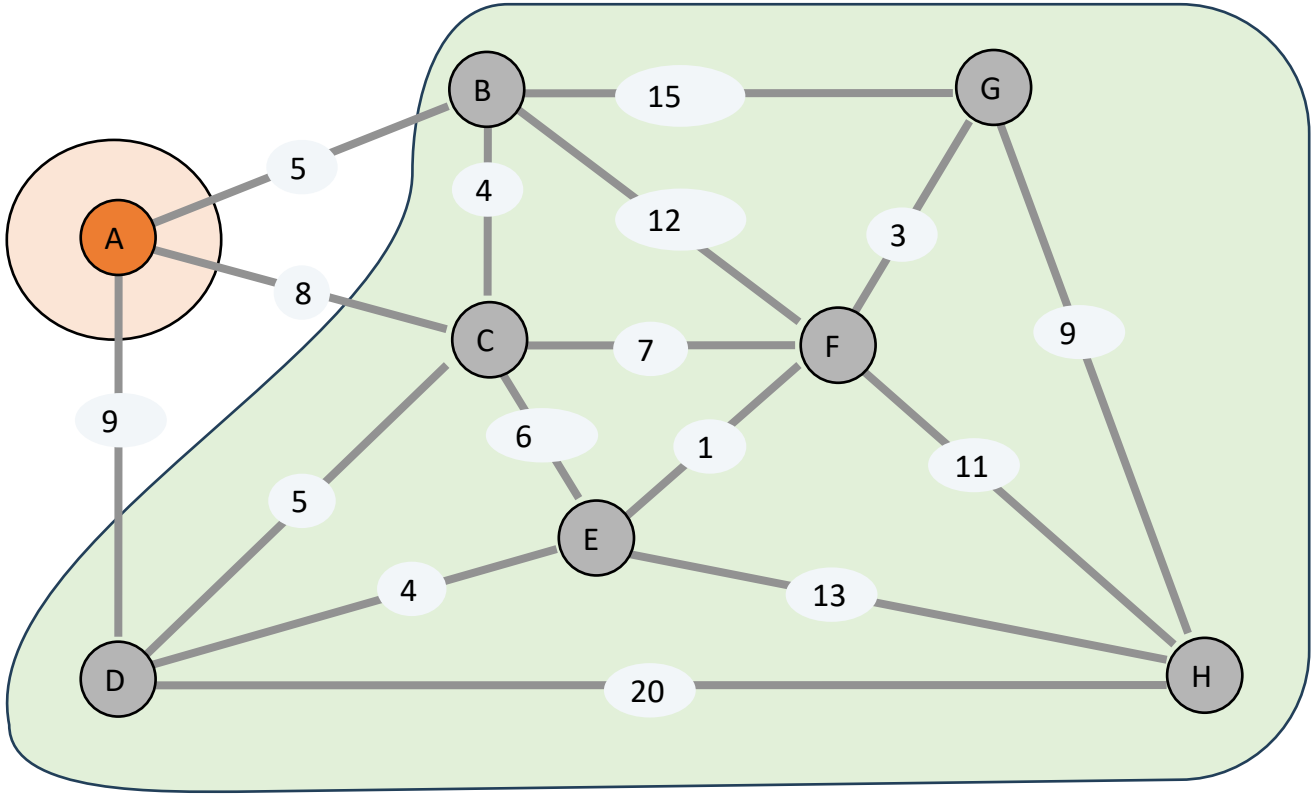

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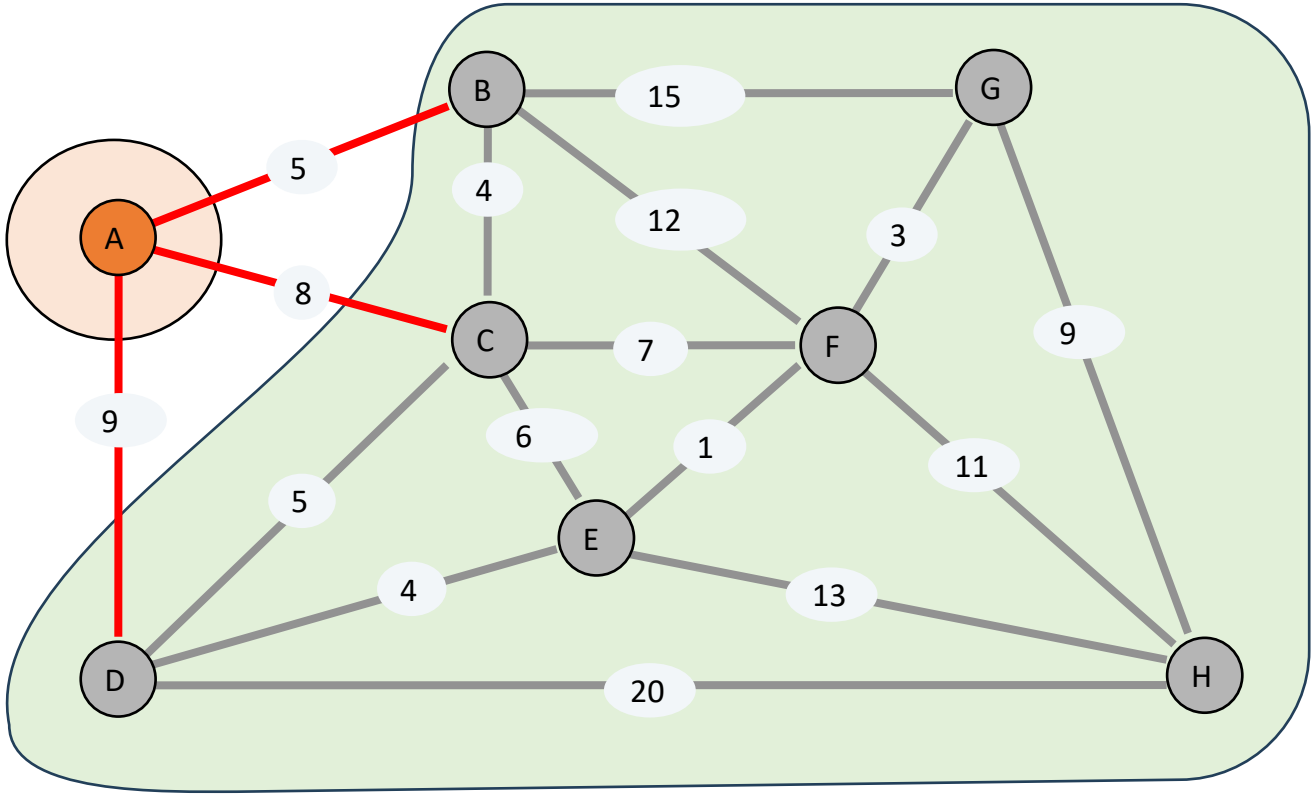

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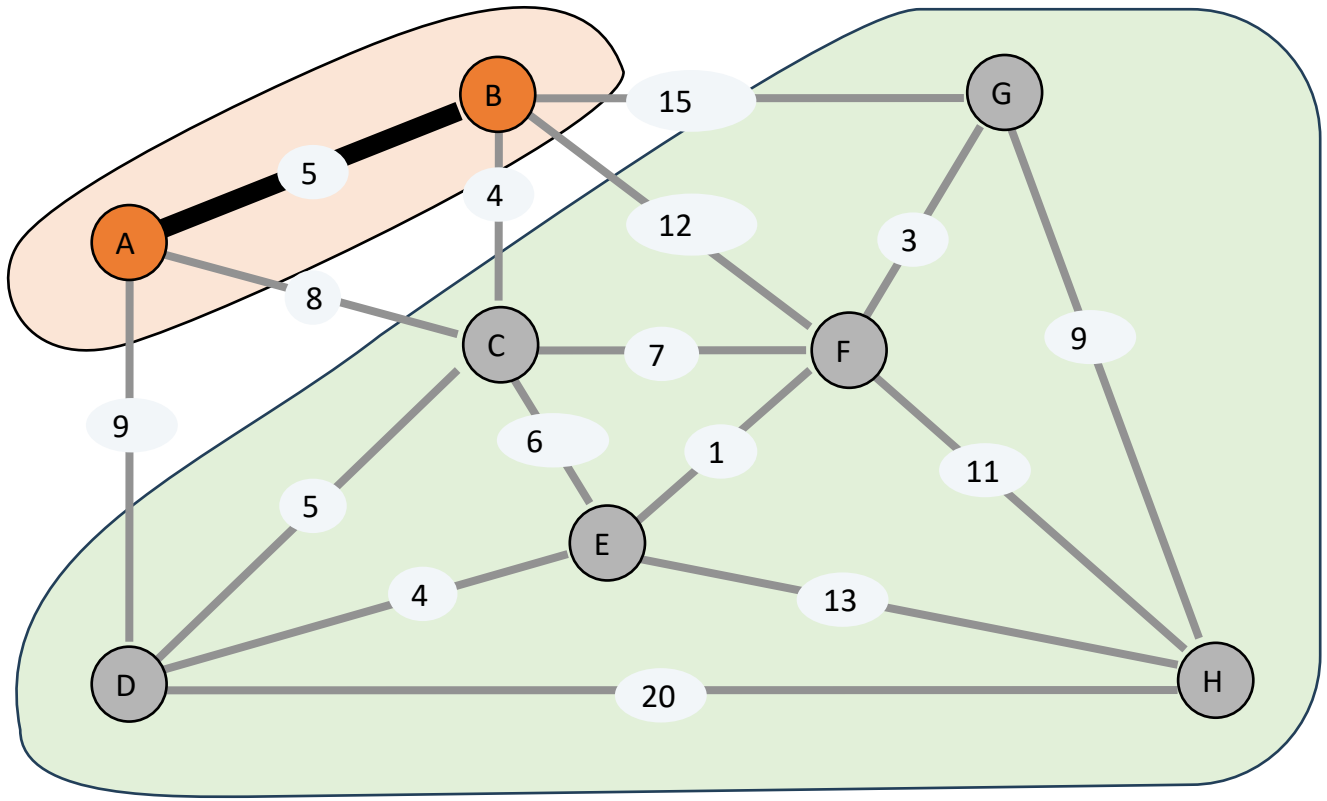

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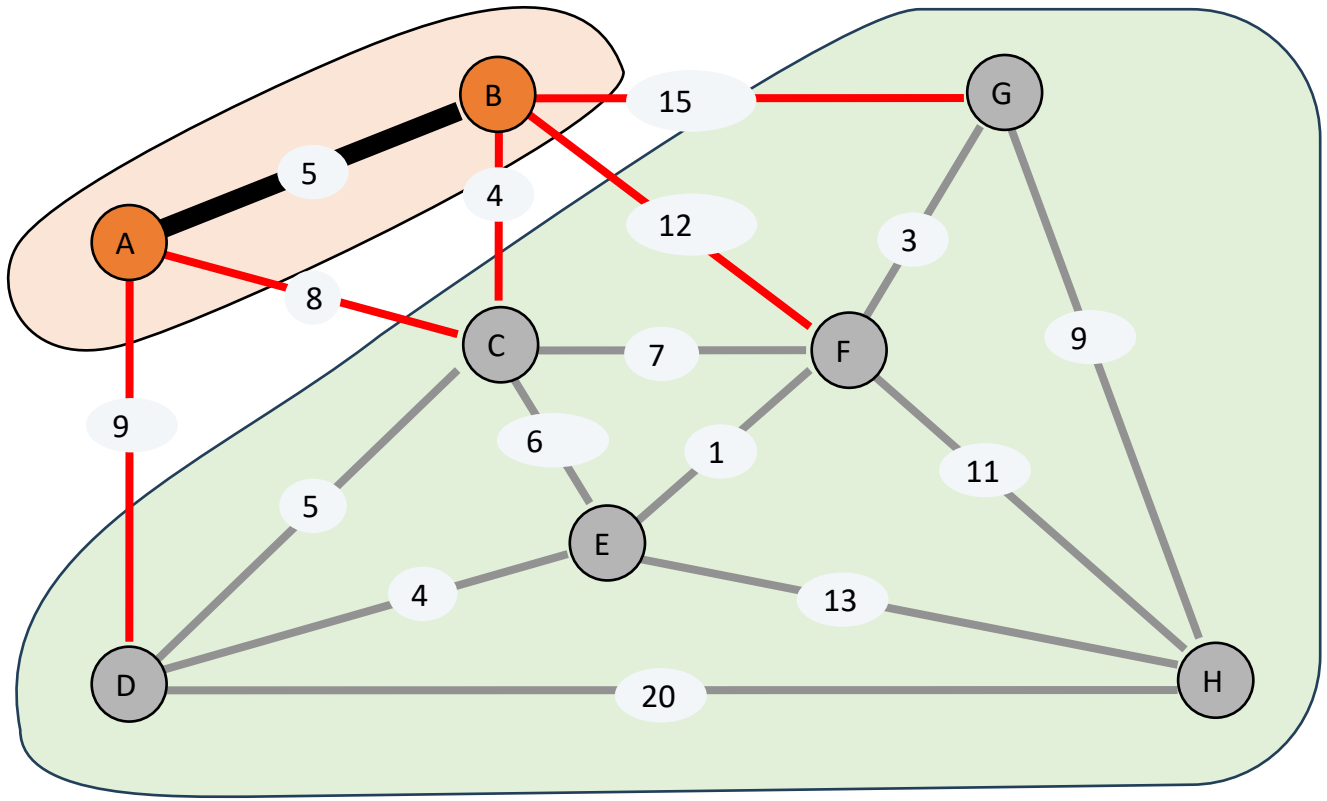

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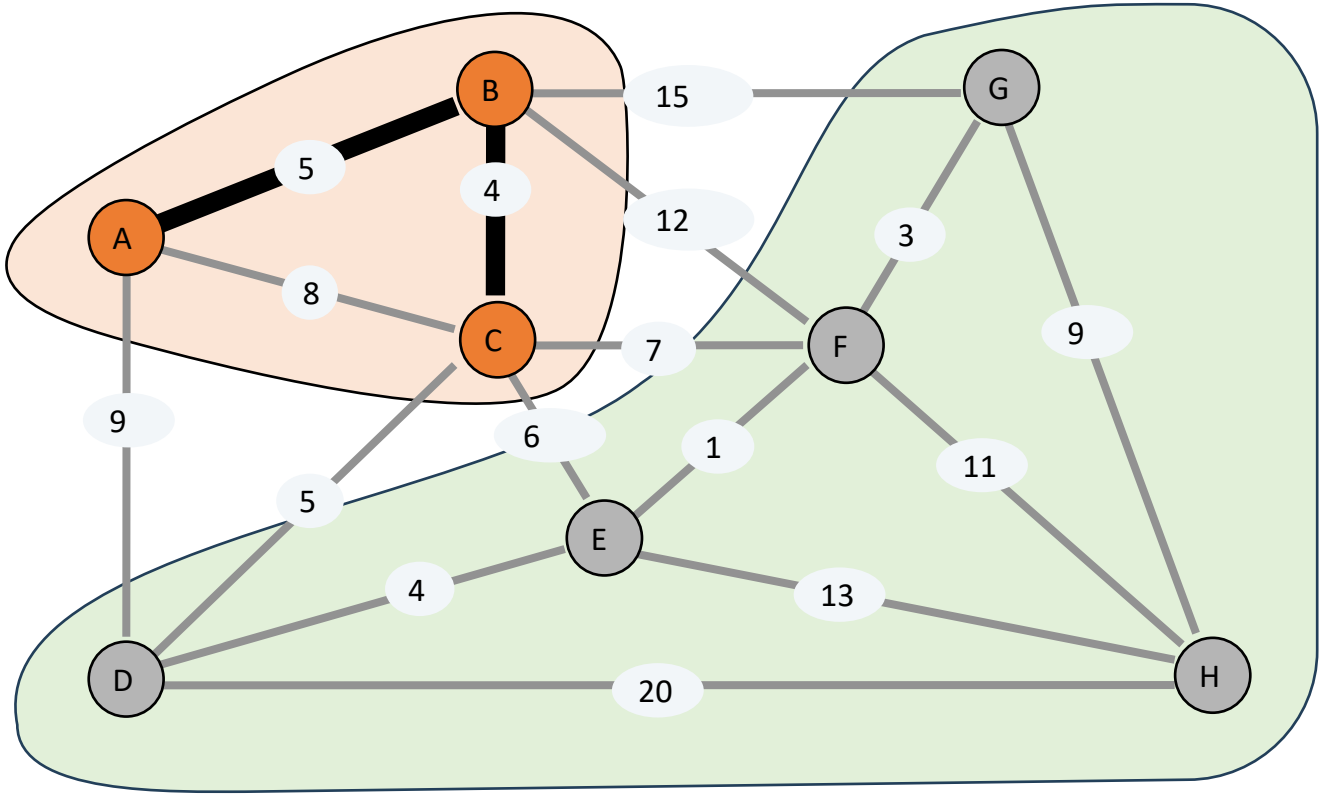

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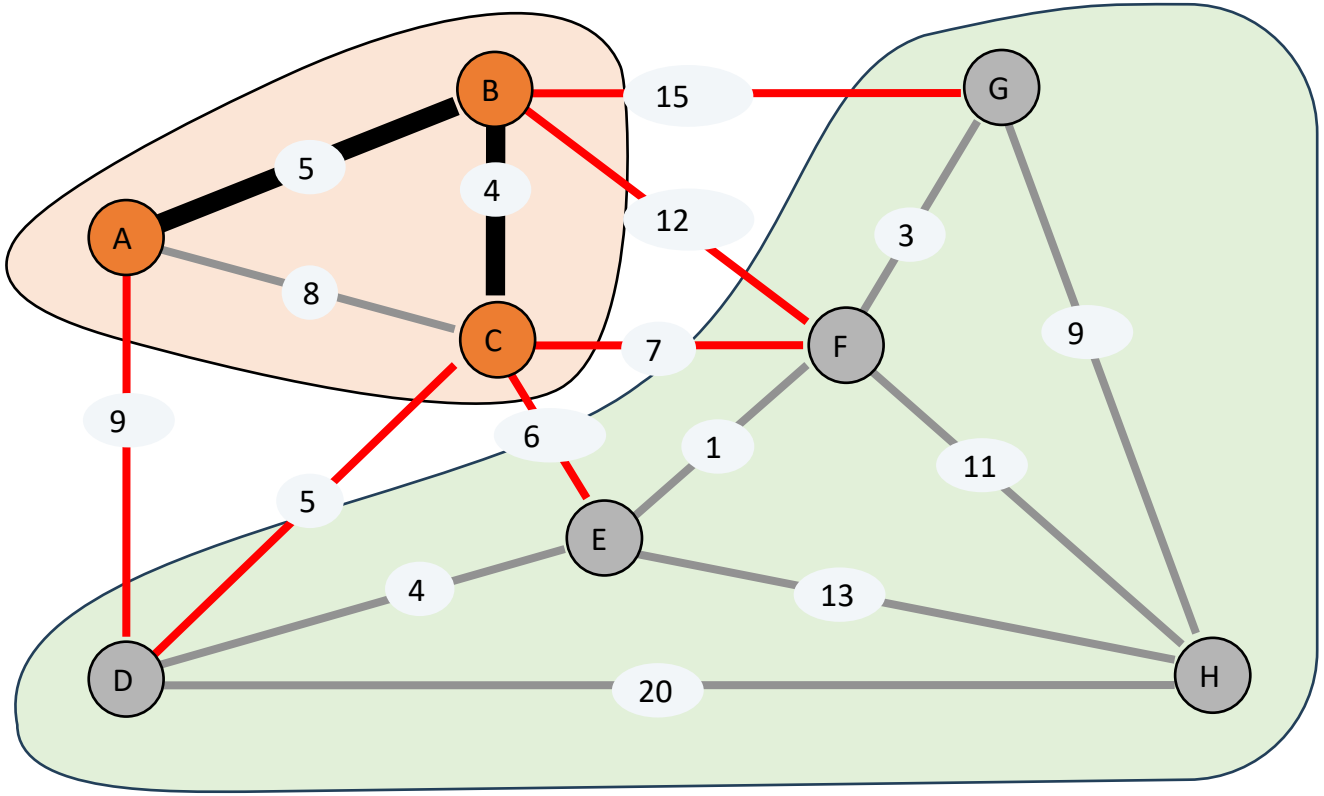

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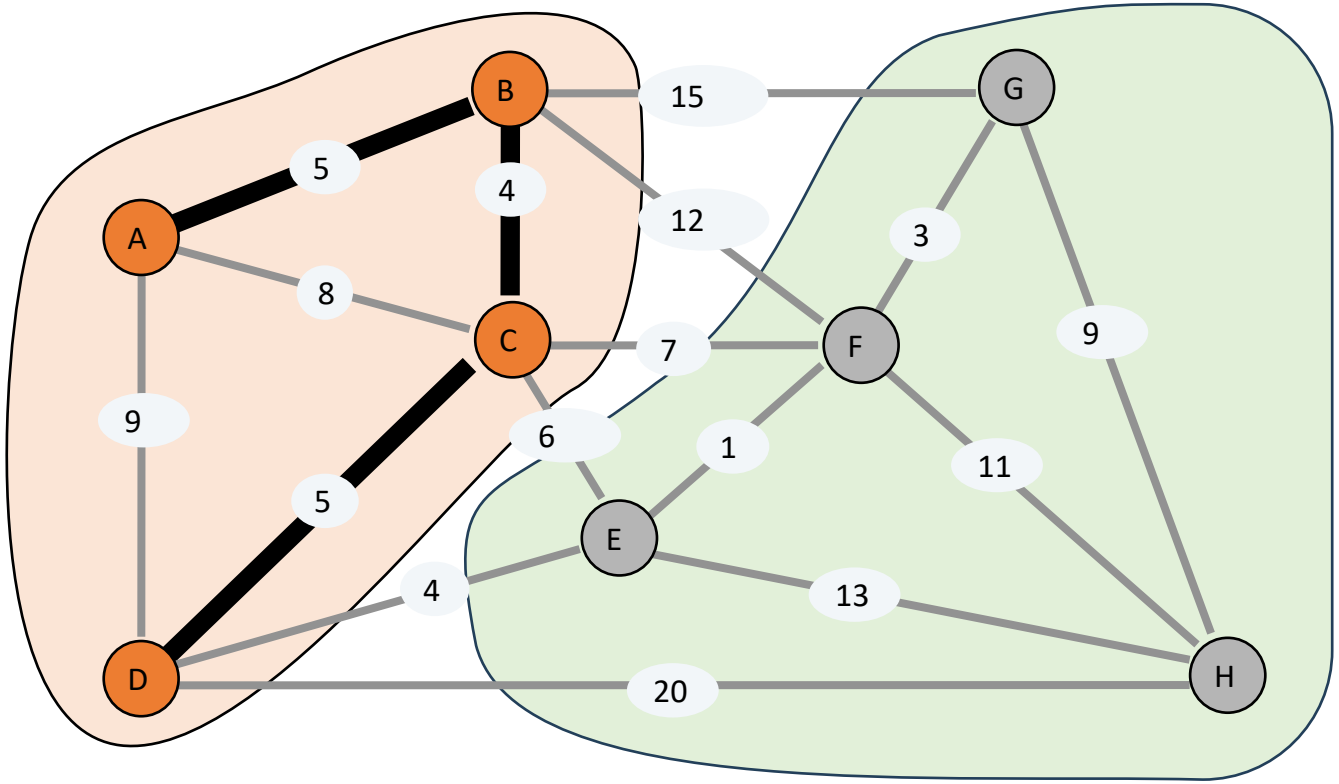

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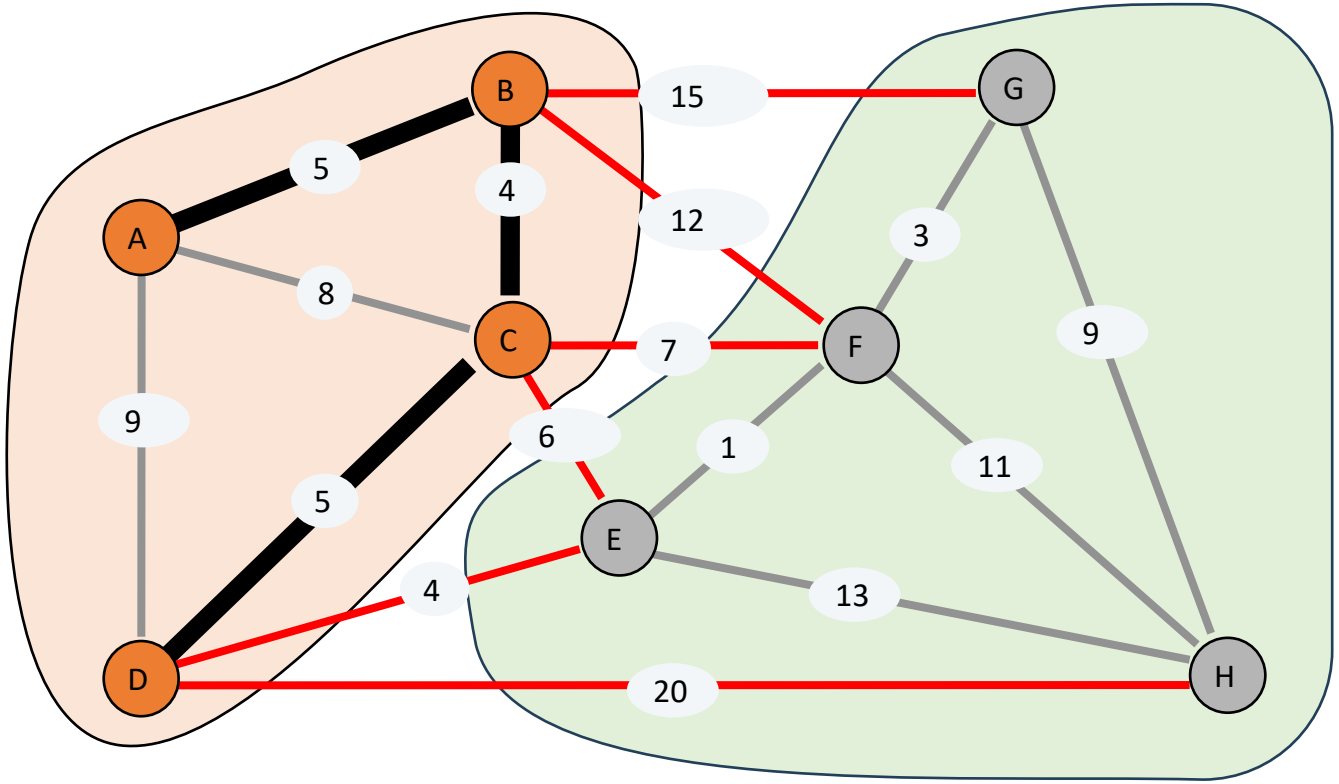

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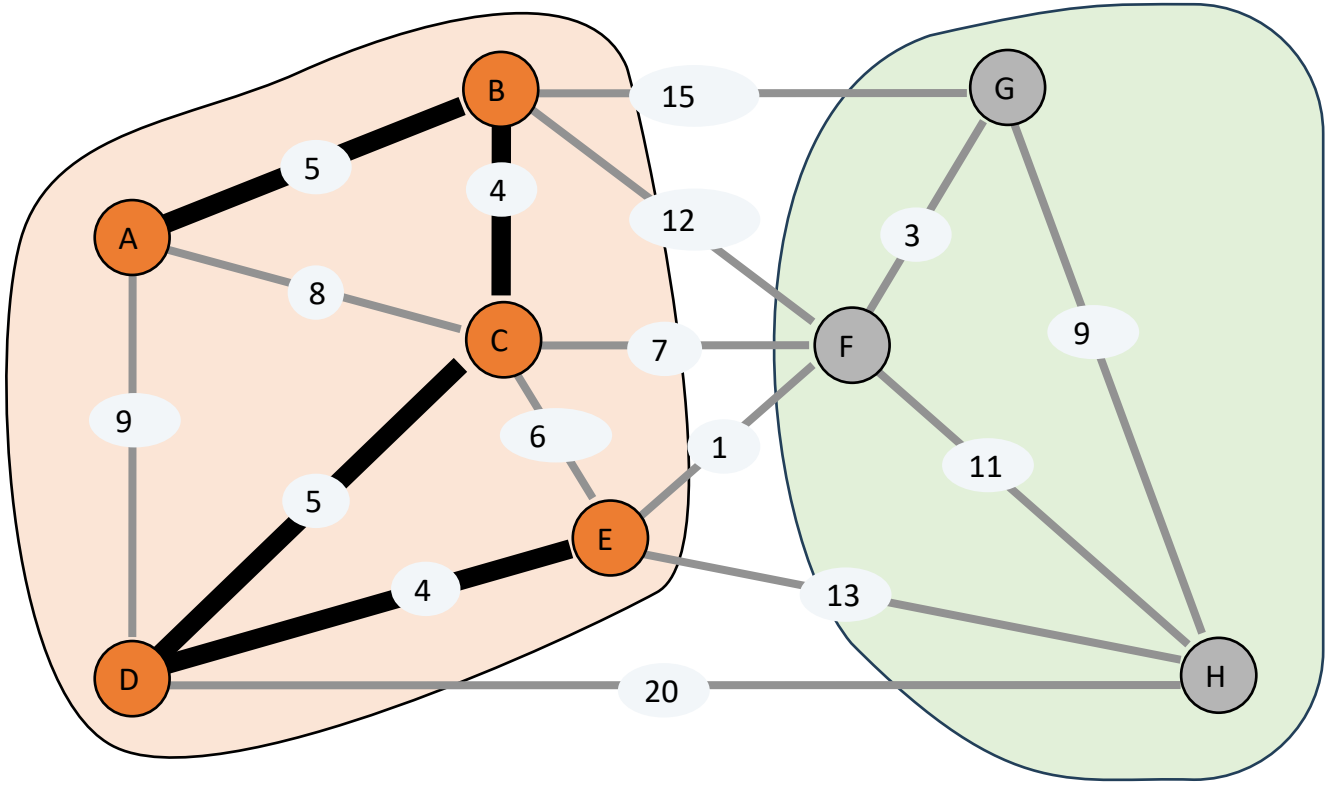

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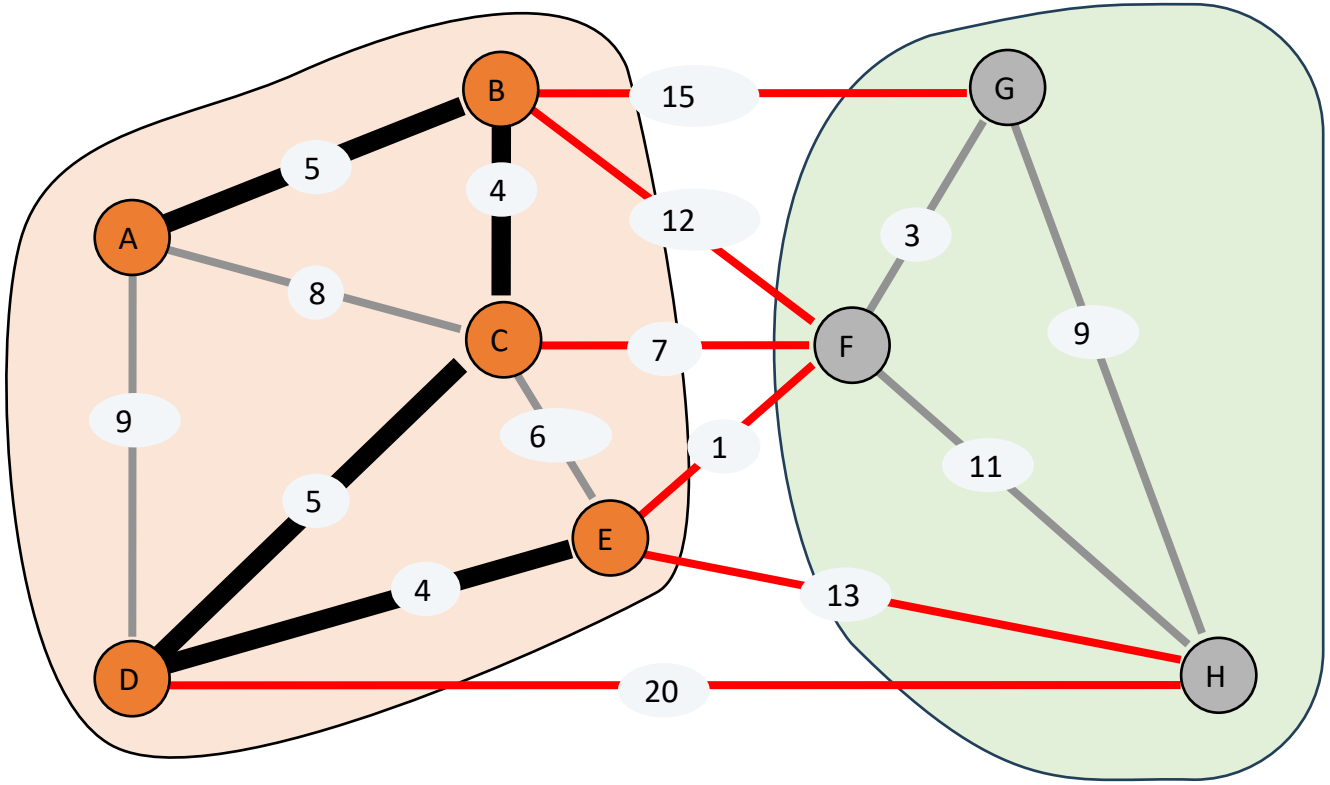

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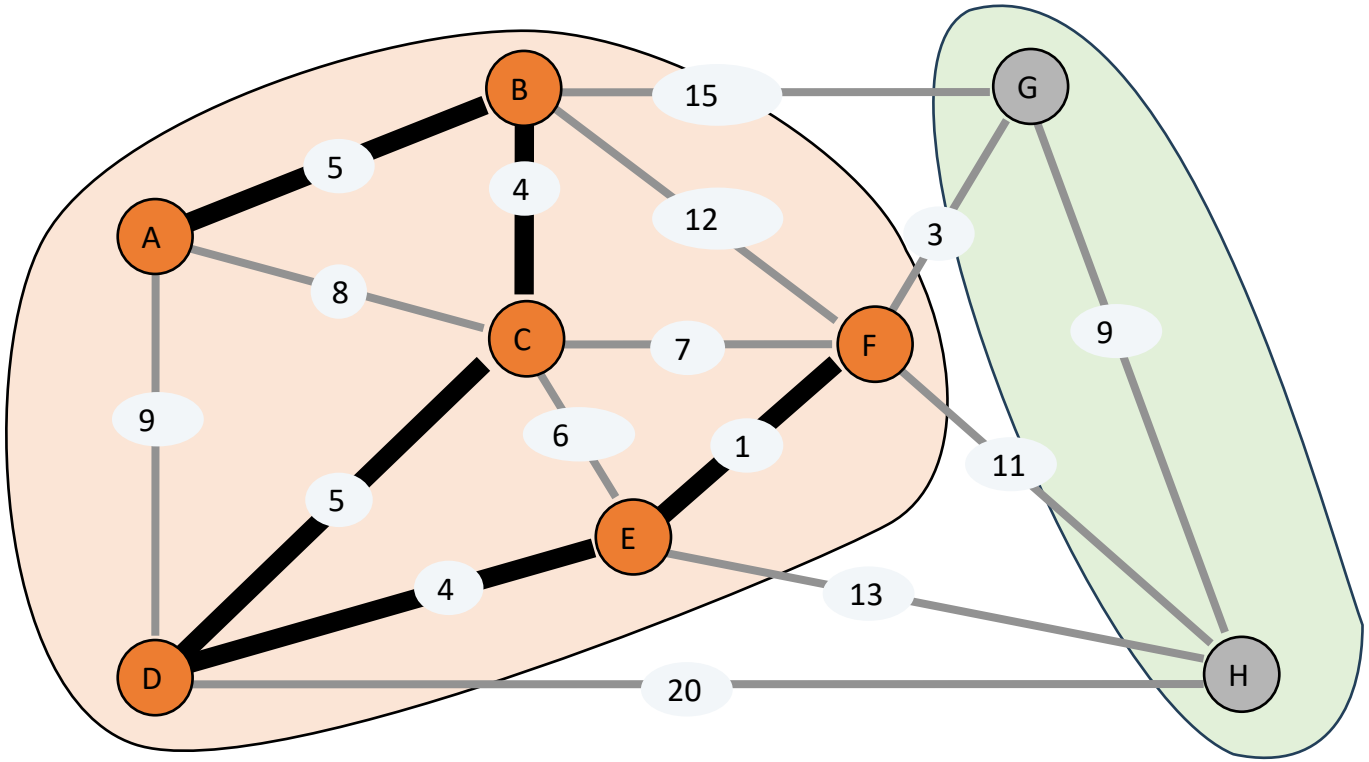

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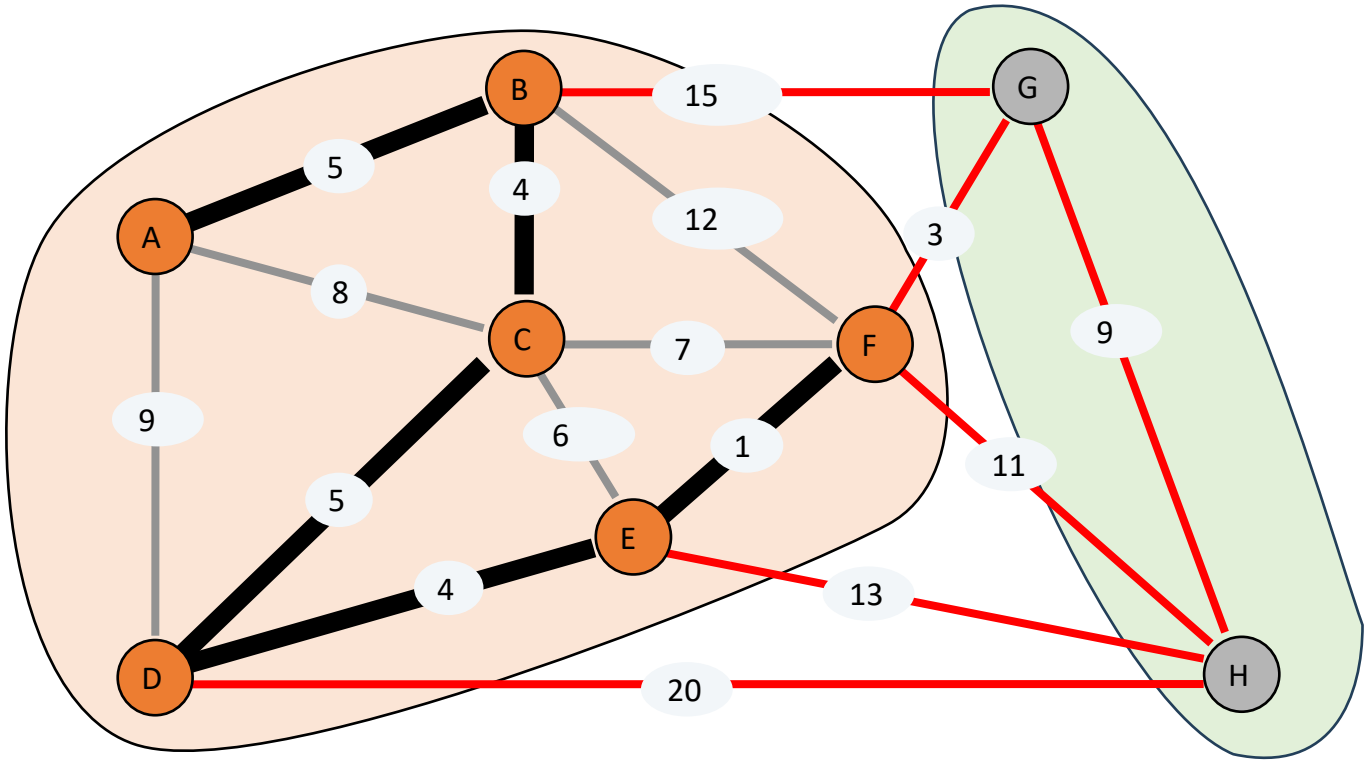

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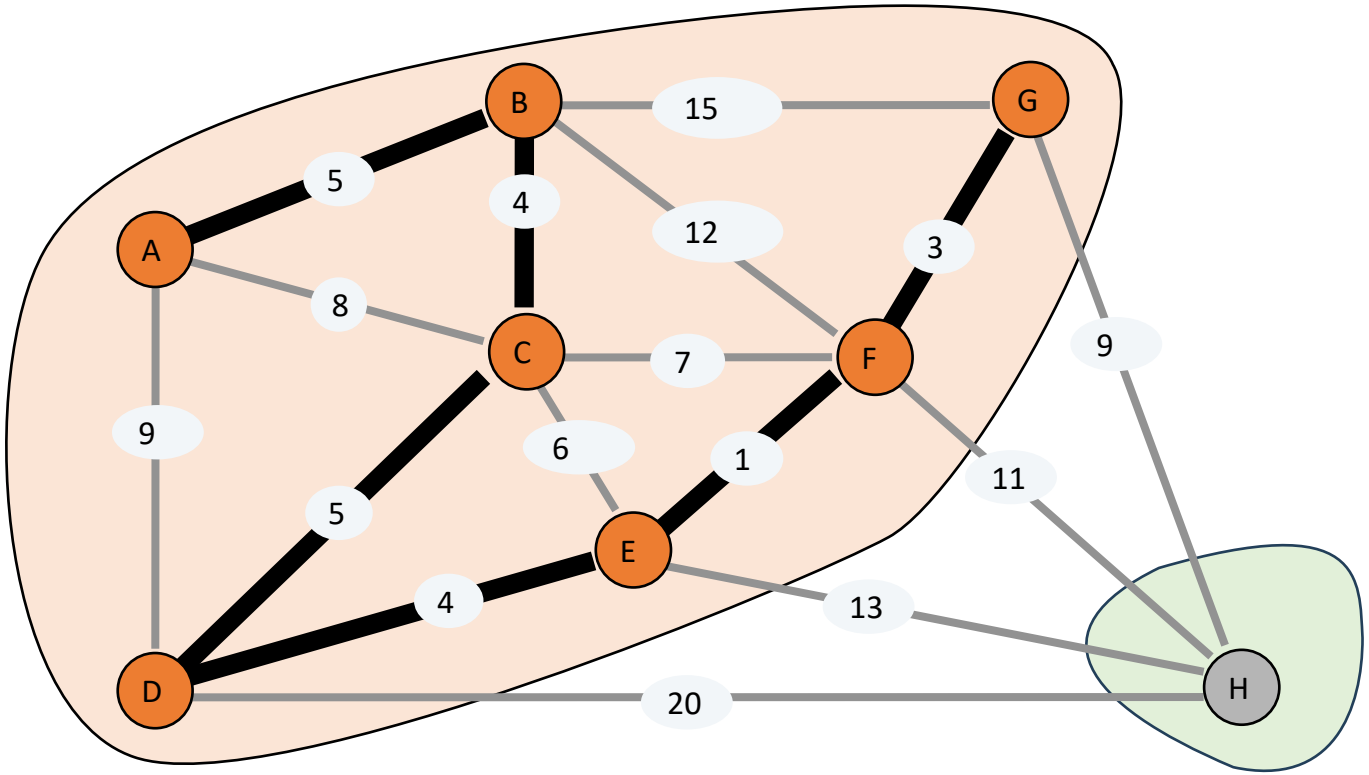

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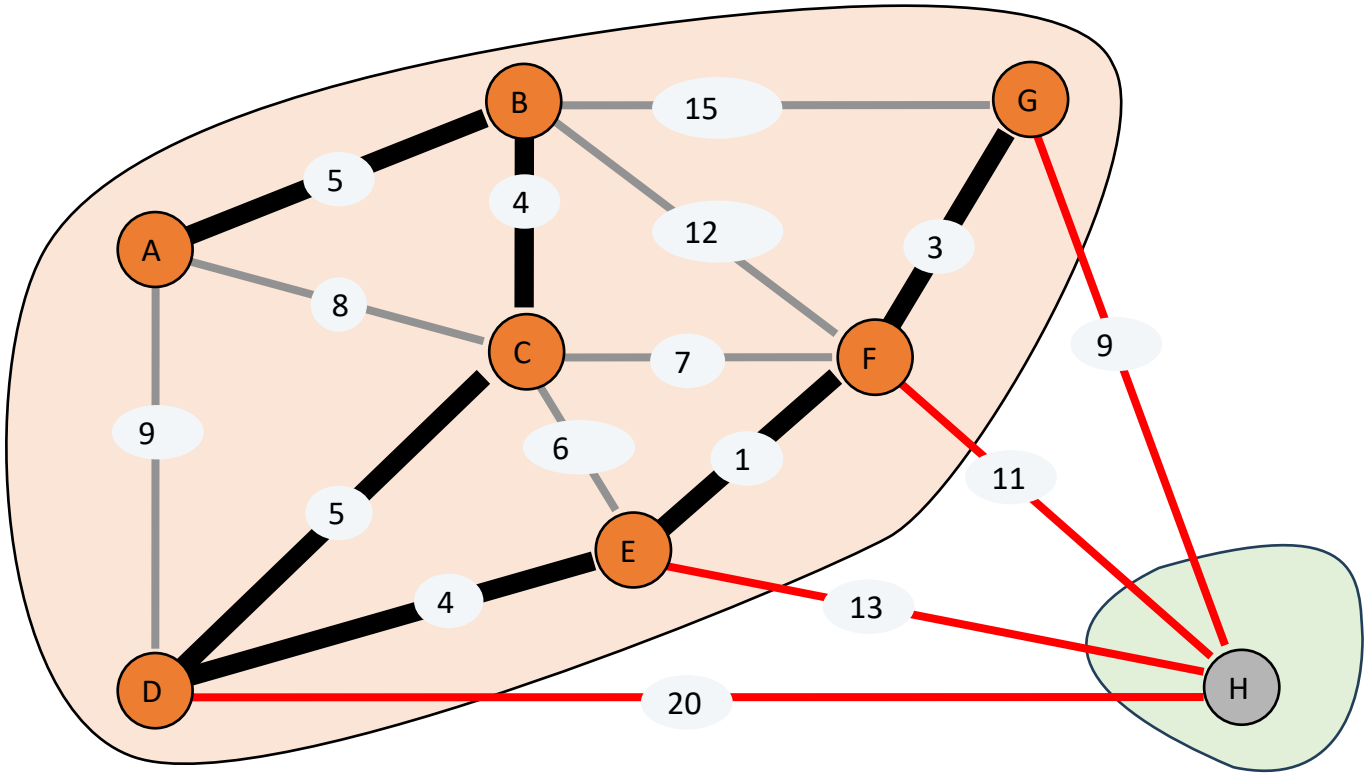

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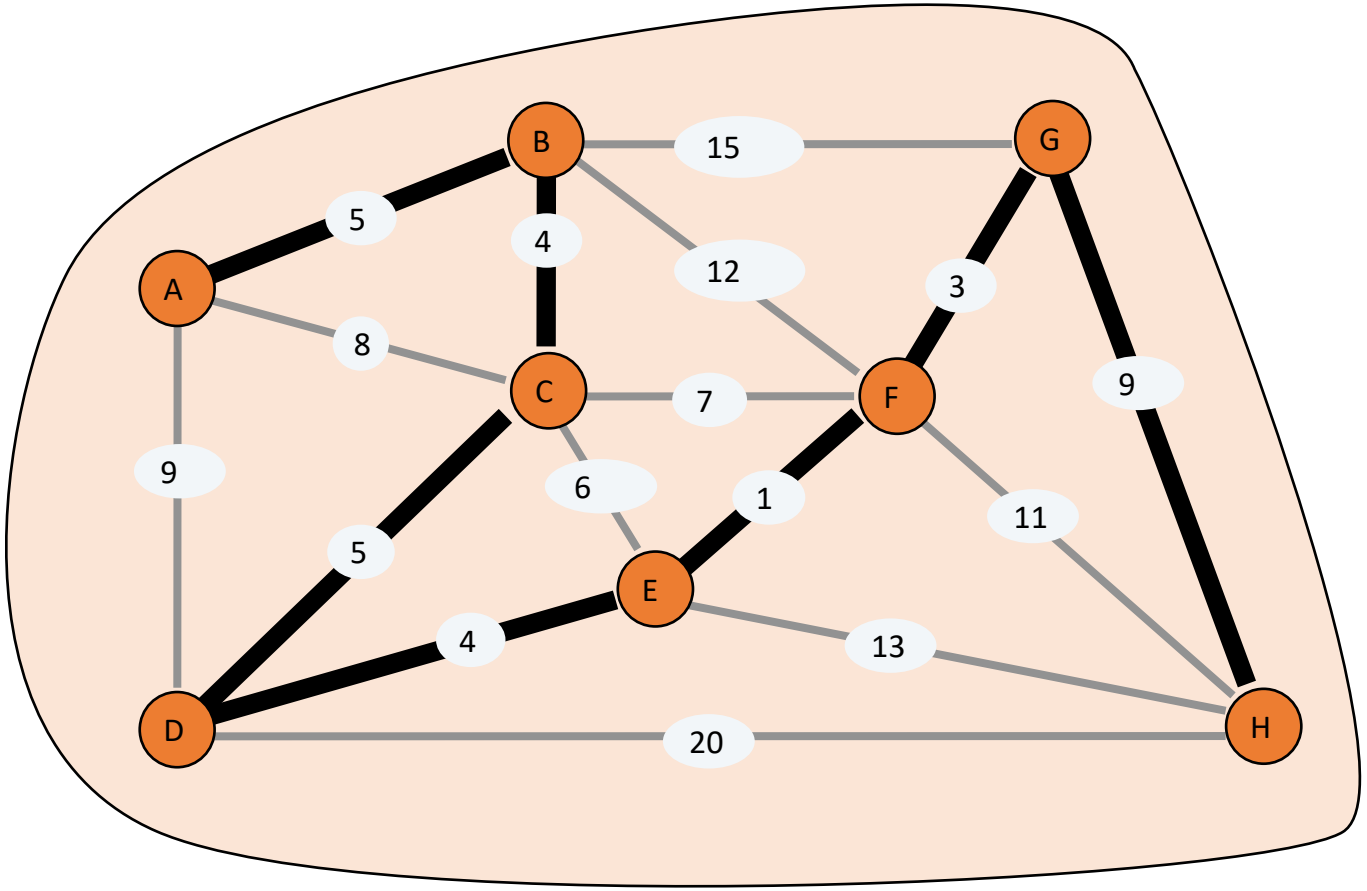
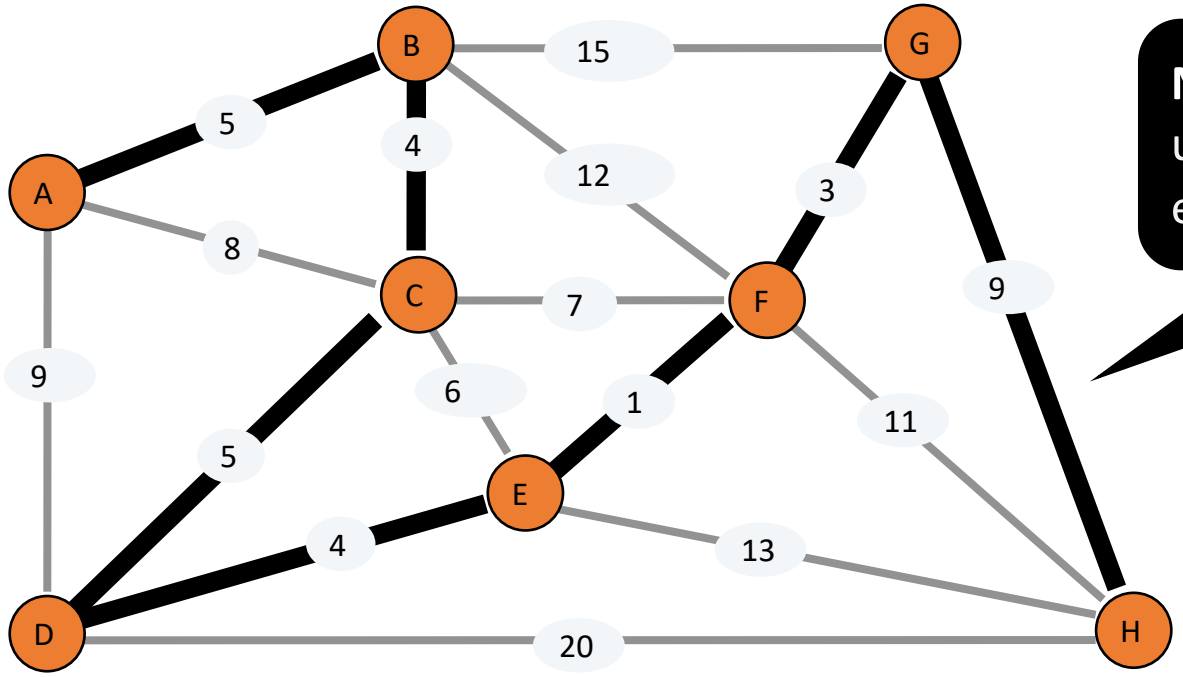


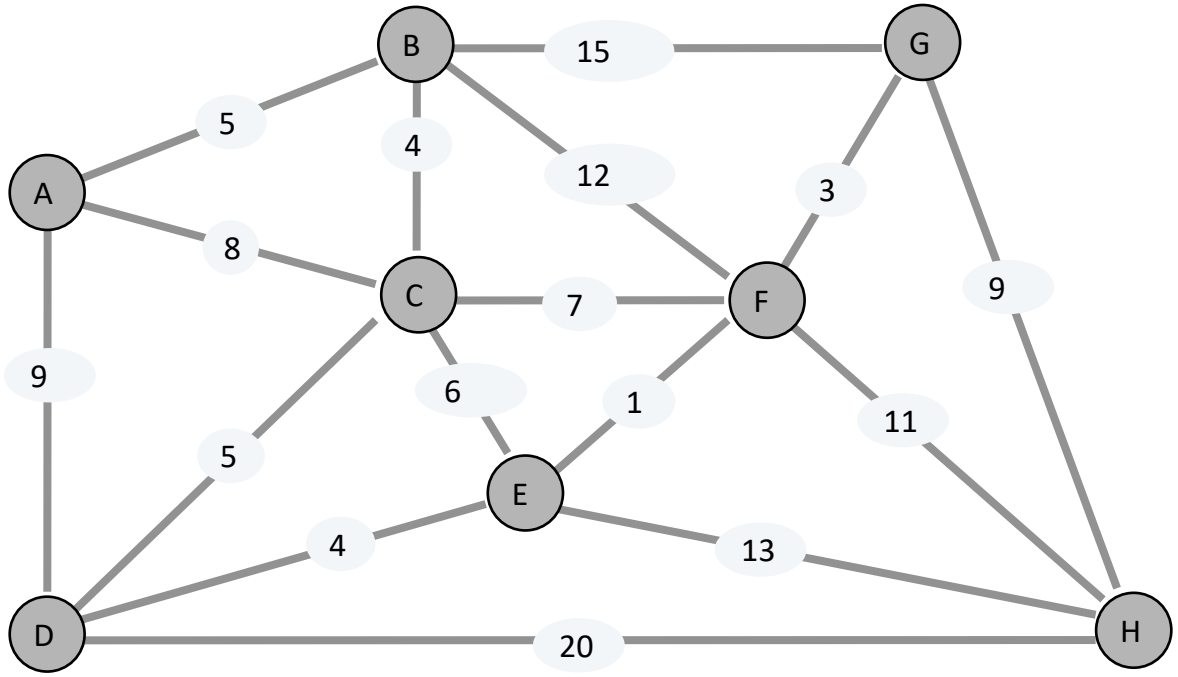
Prim's algorithm Time Complexity (Same as Dijkstra)



Note: MST will always be unique for each graph given edge **weights are unique**

Kruskal's algorithm to find MST

- Consider edges in ascending order of weight



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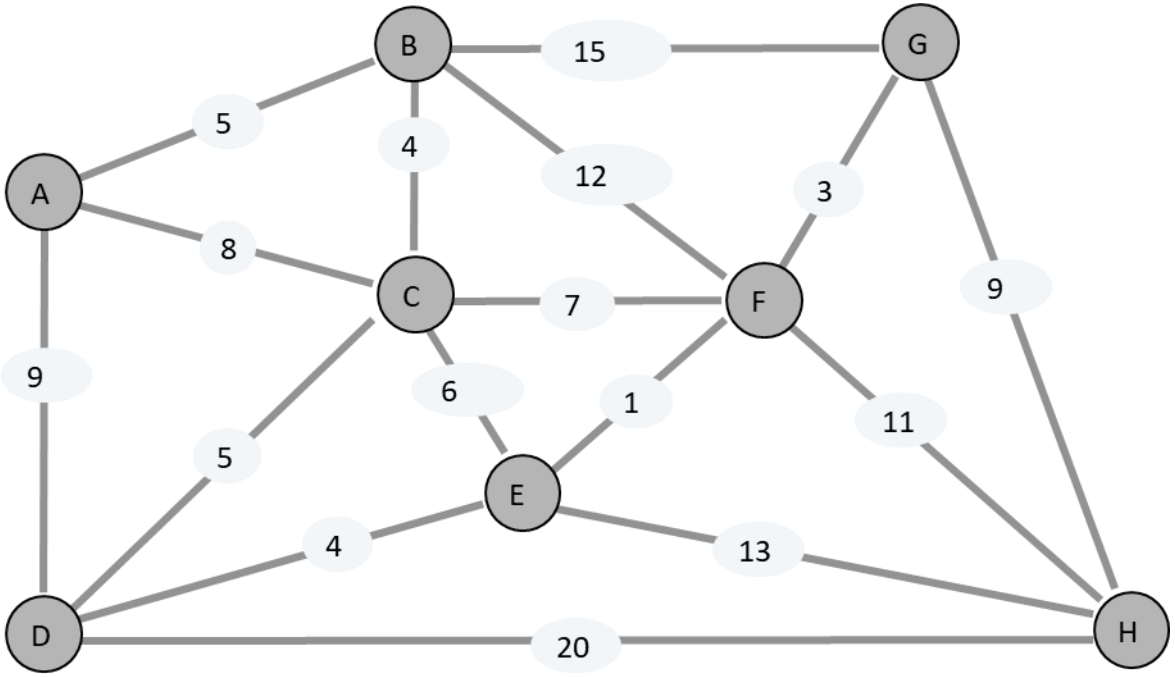

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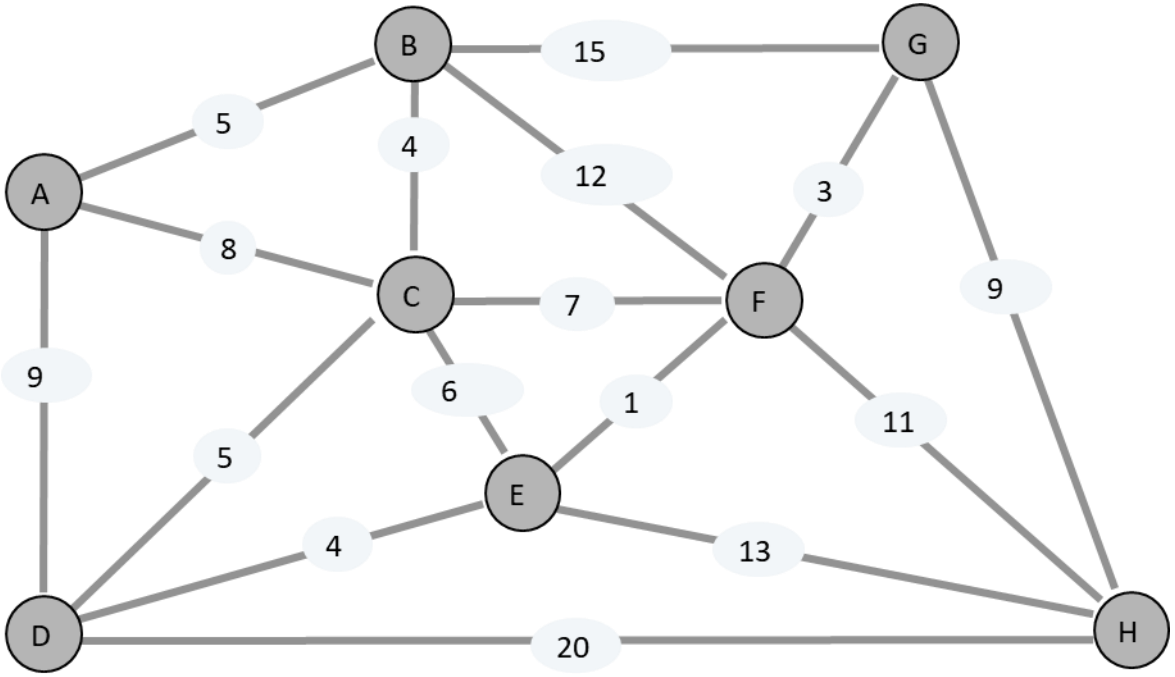
Kruskal's algorithm to find MST

- Consider edges in ascending order of weight
- Add to e to MST unless it would create a cycle



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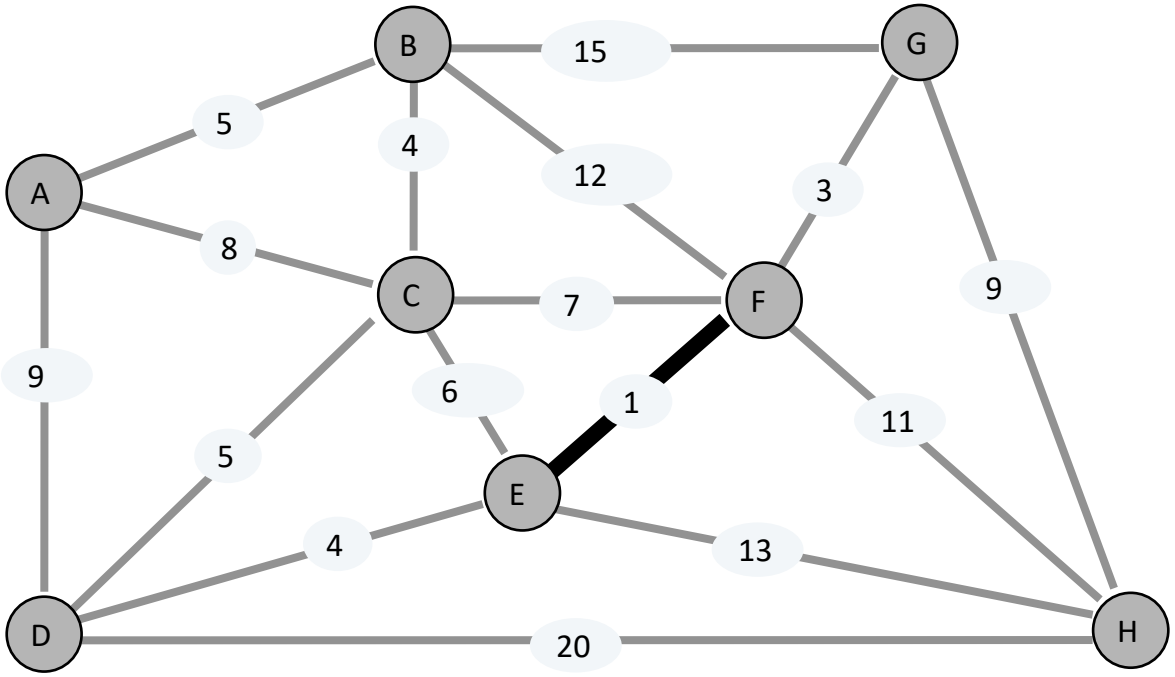

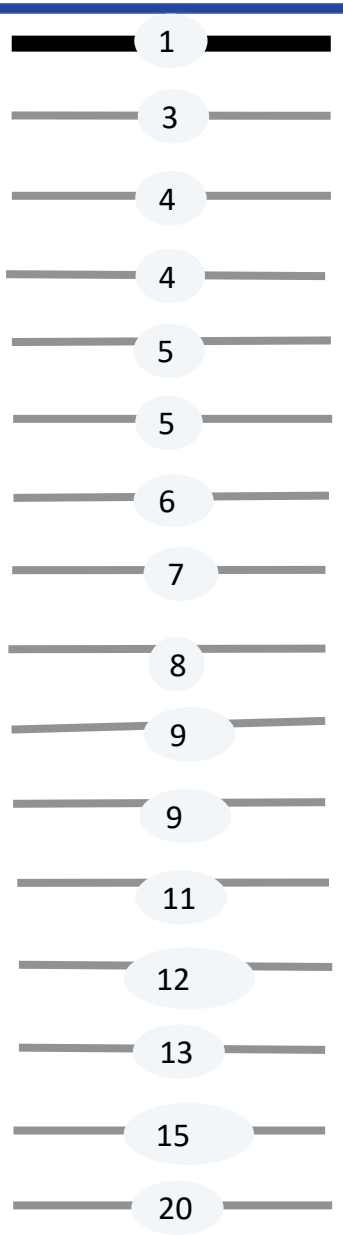

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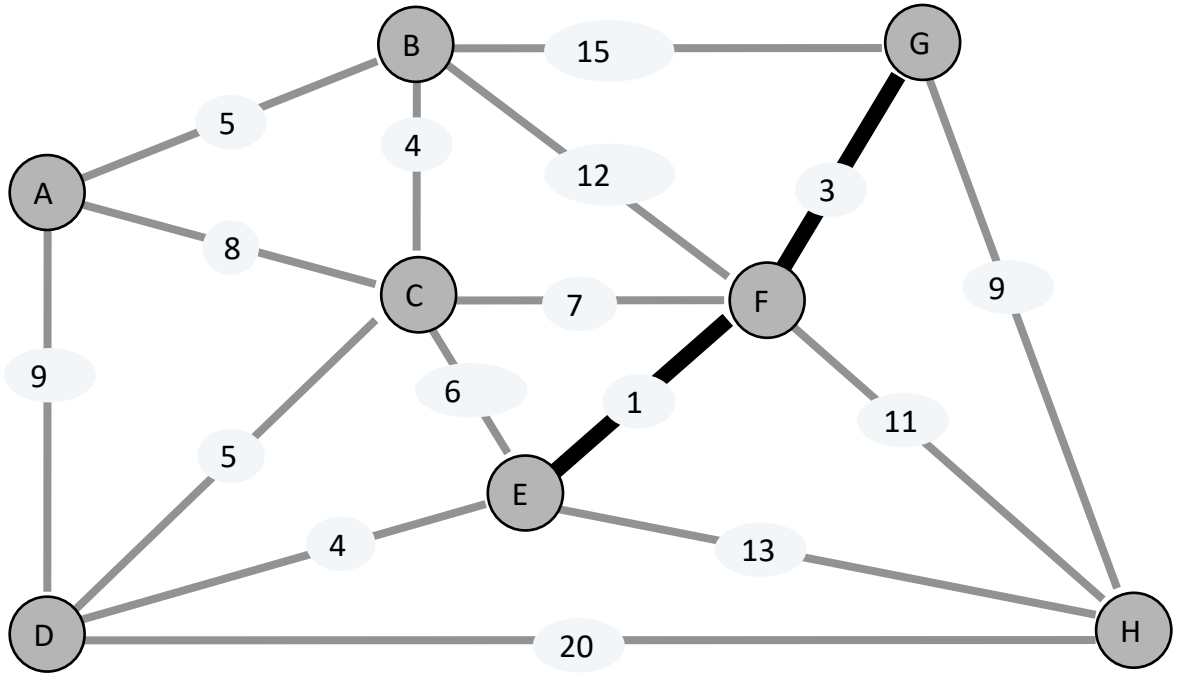

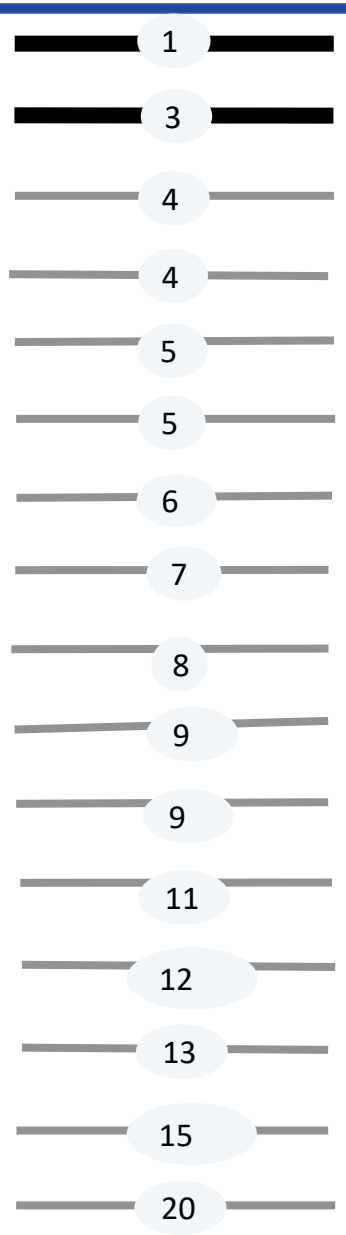

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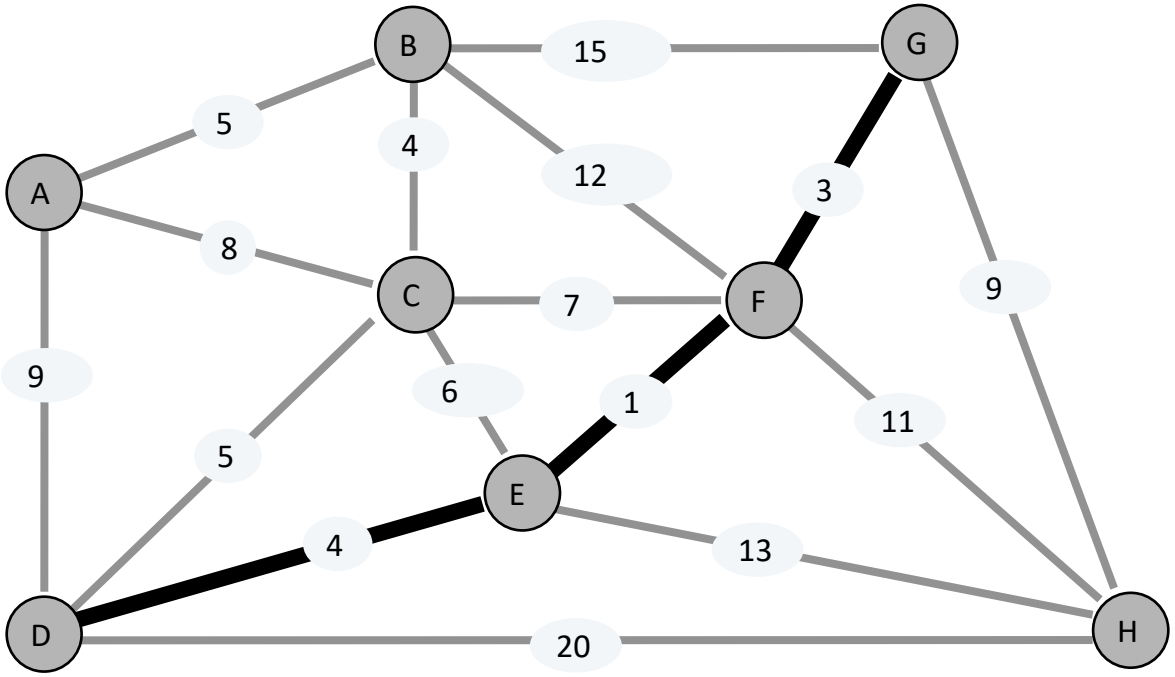

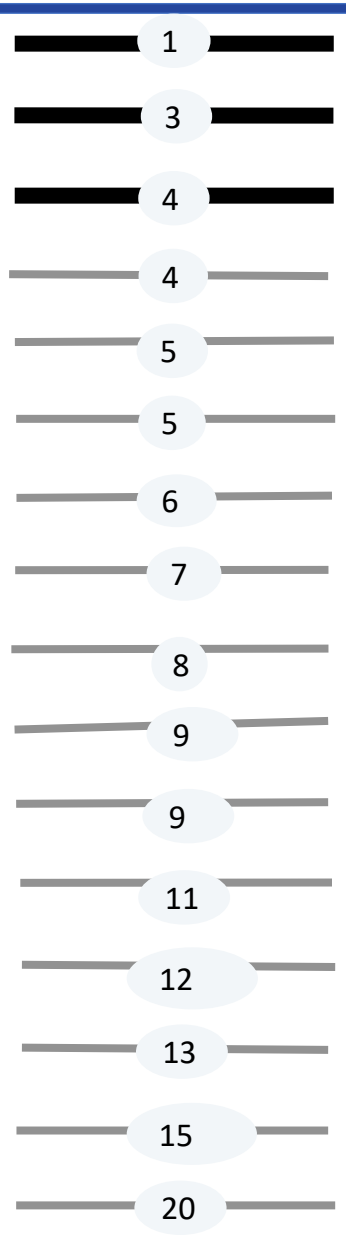

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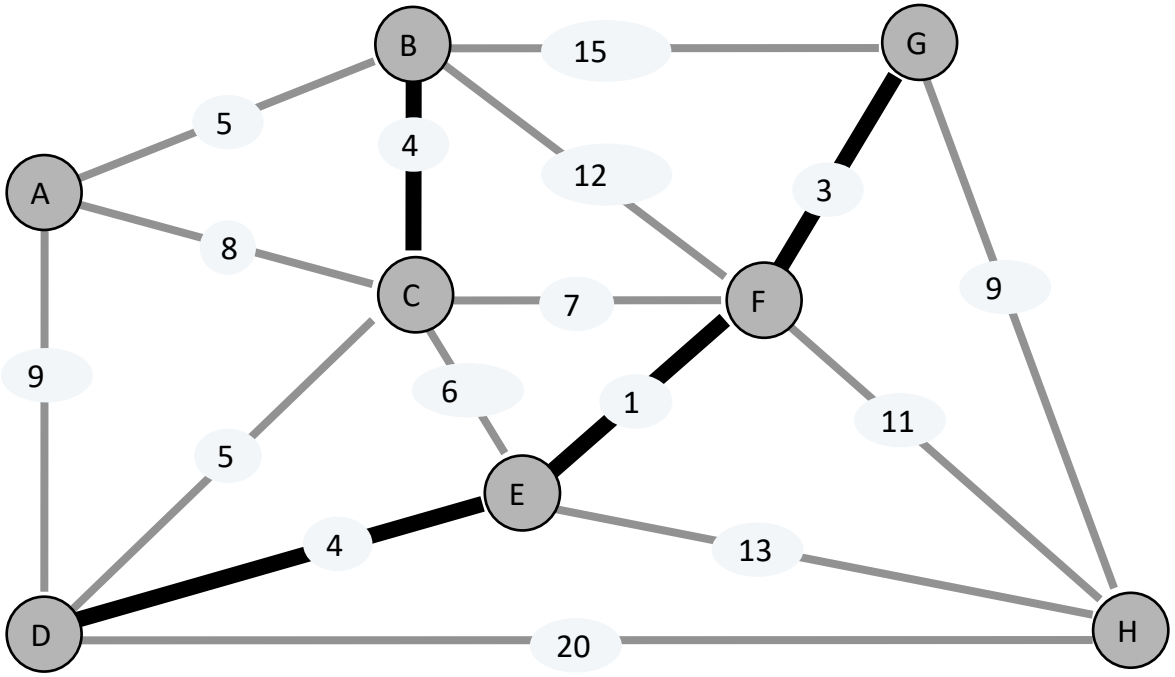

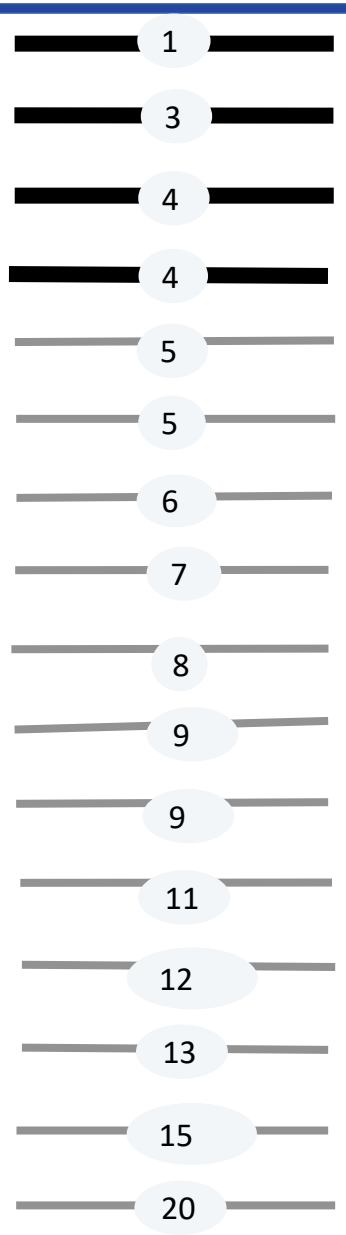

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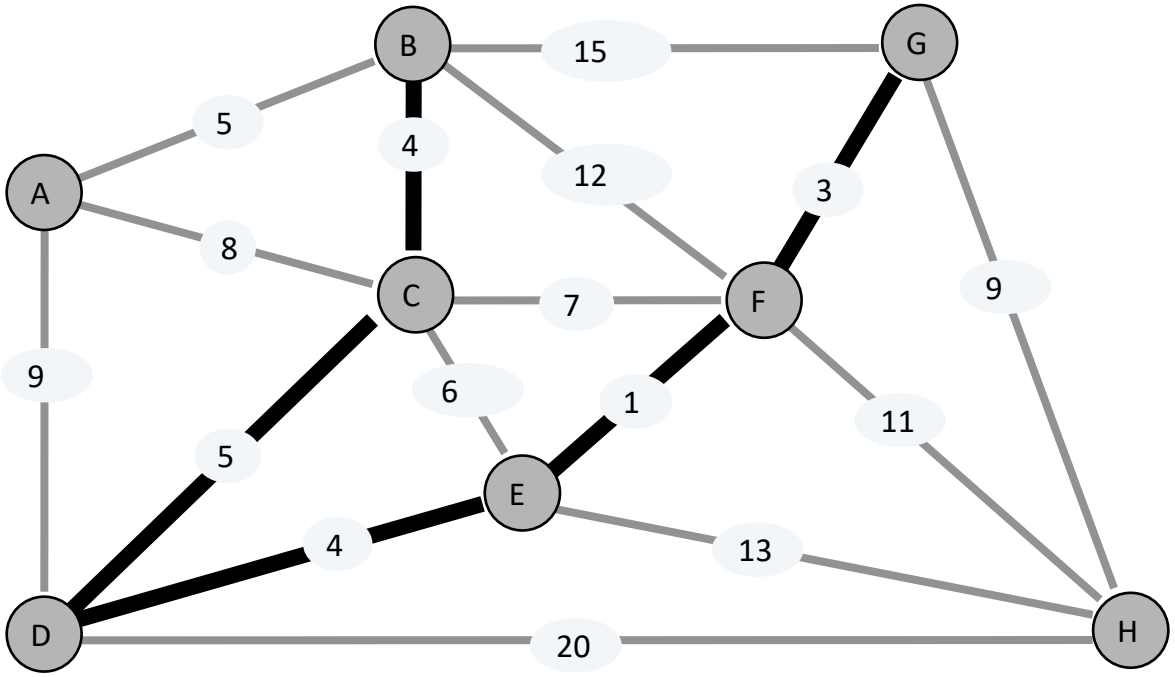
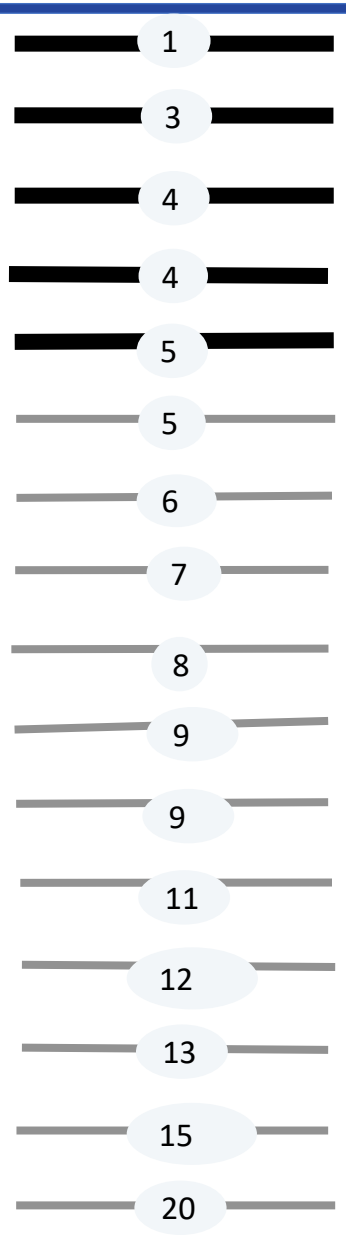
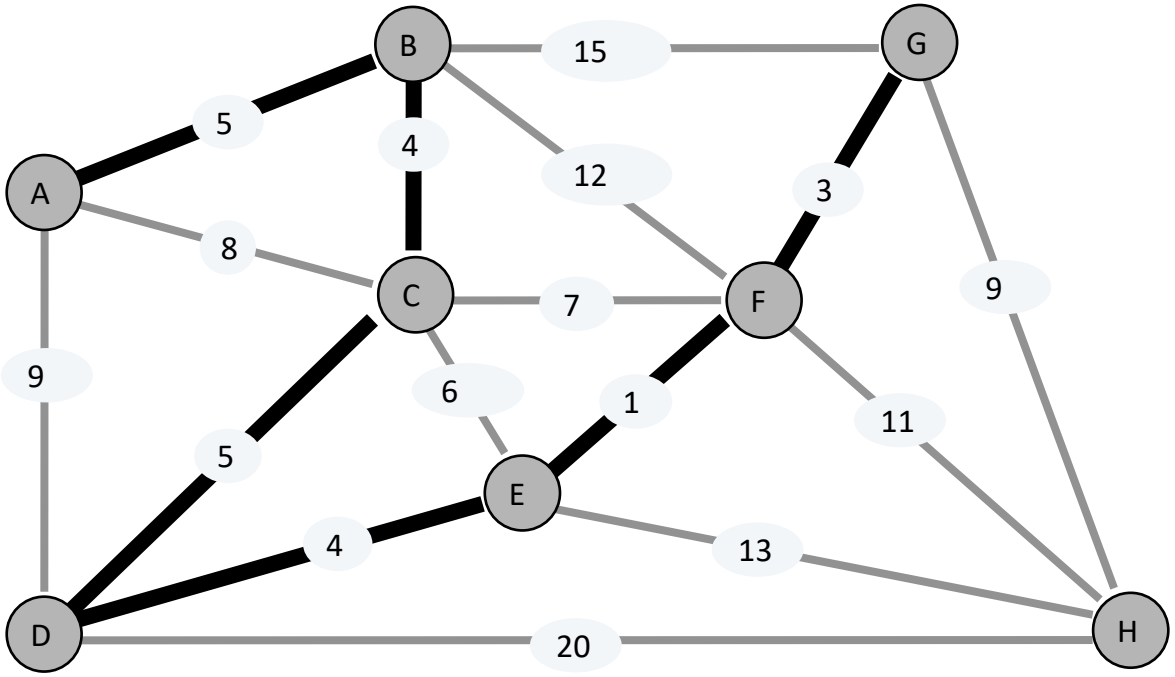
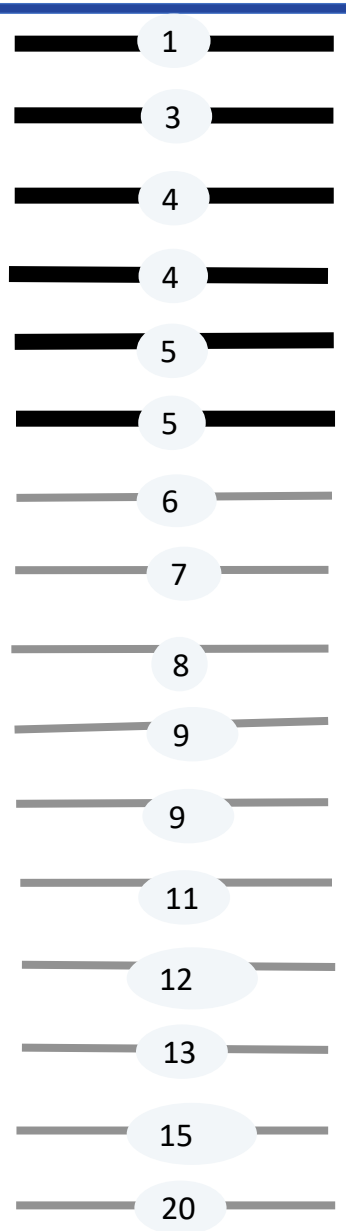




Kruskal's algorithm to find MST





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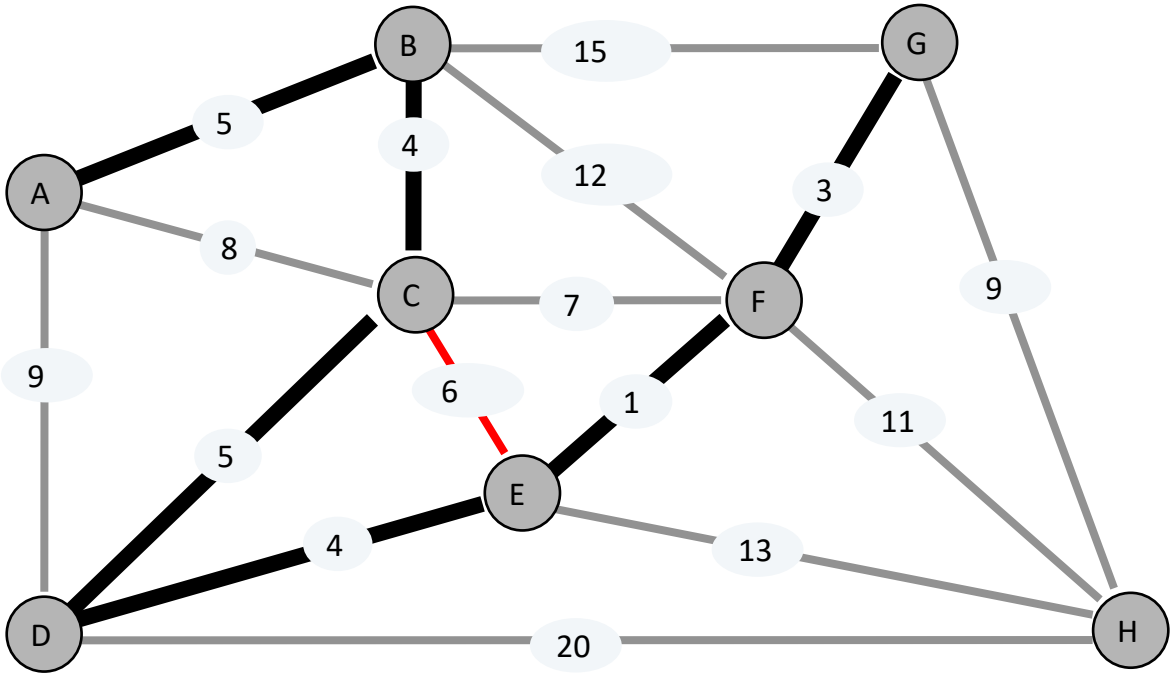

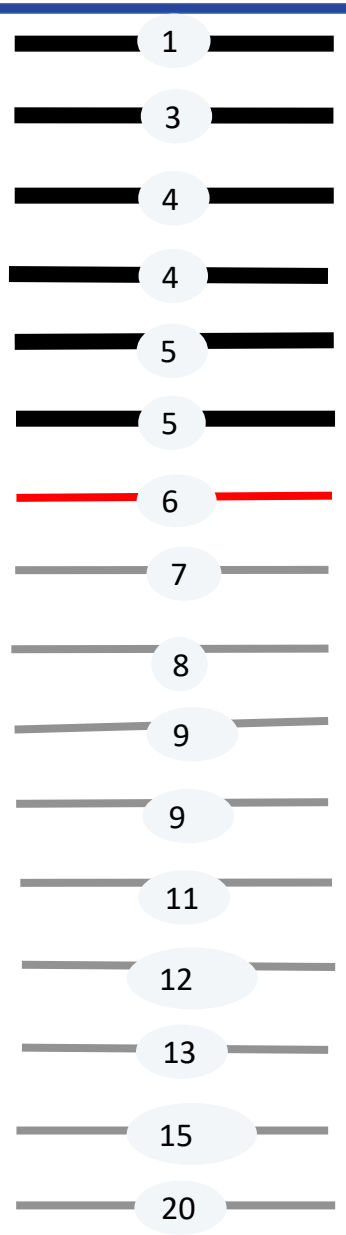

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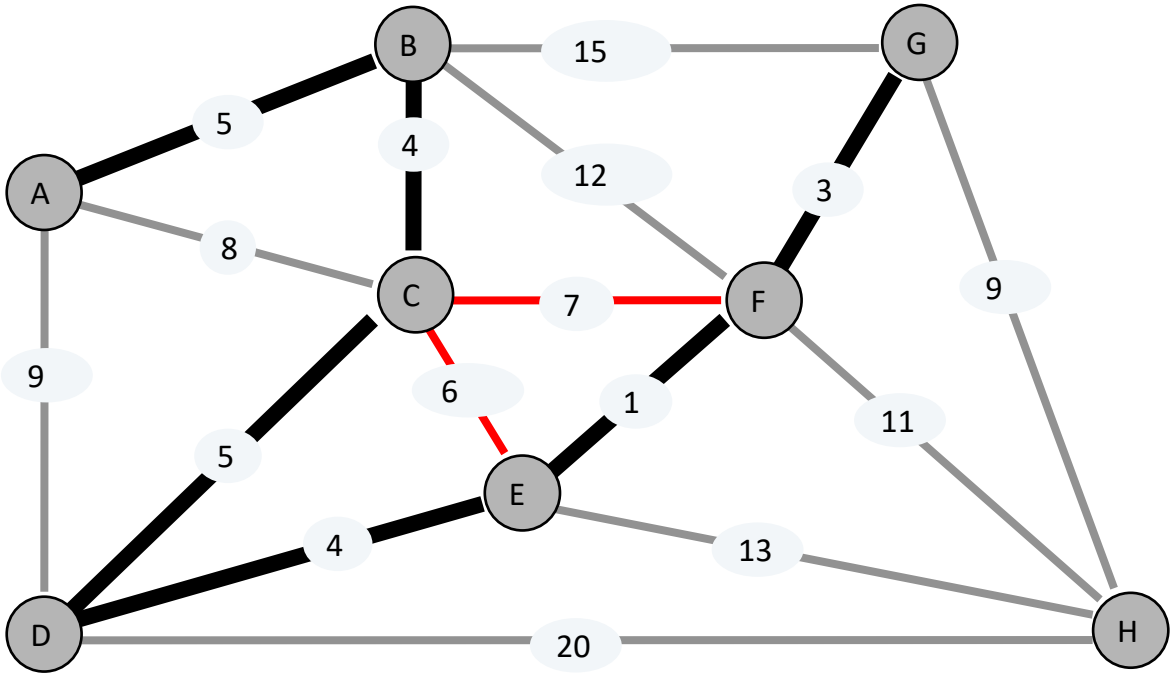
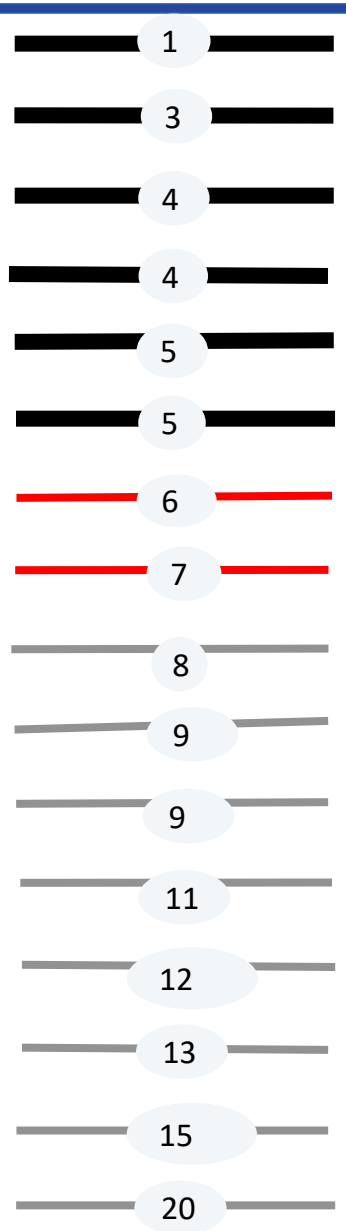
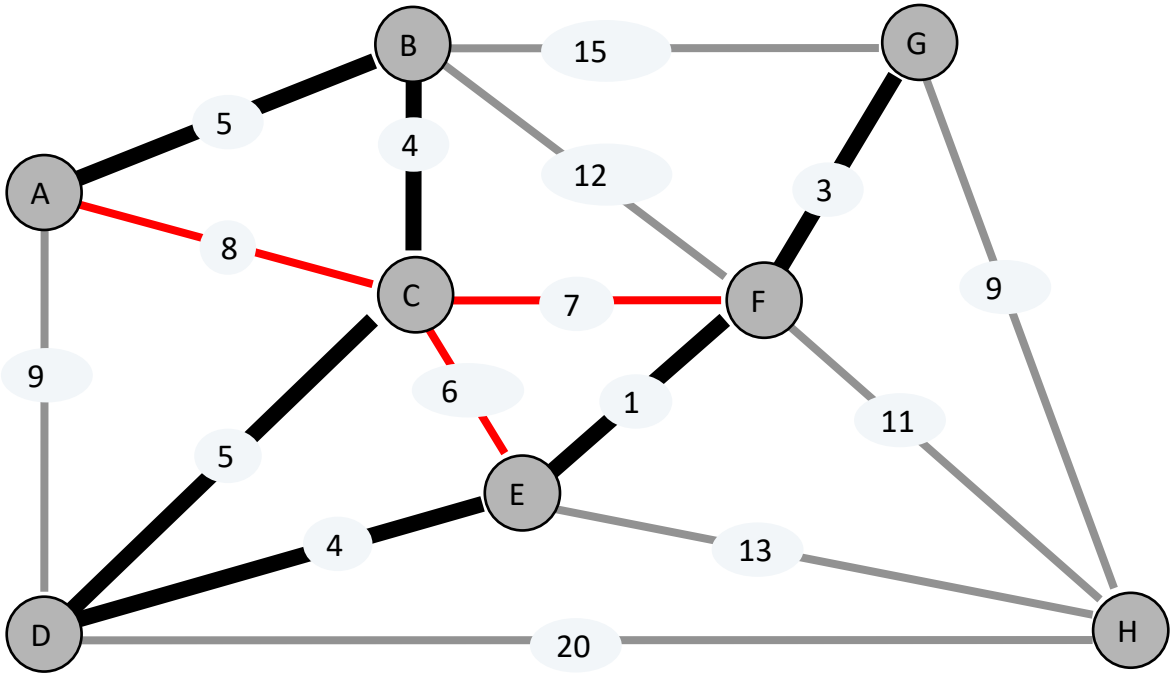
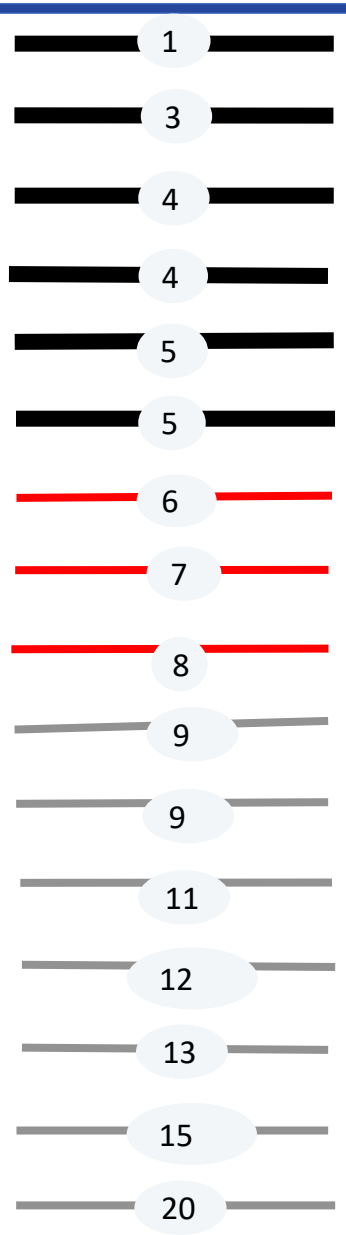




Kruskal's algorithm to find MST





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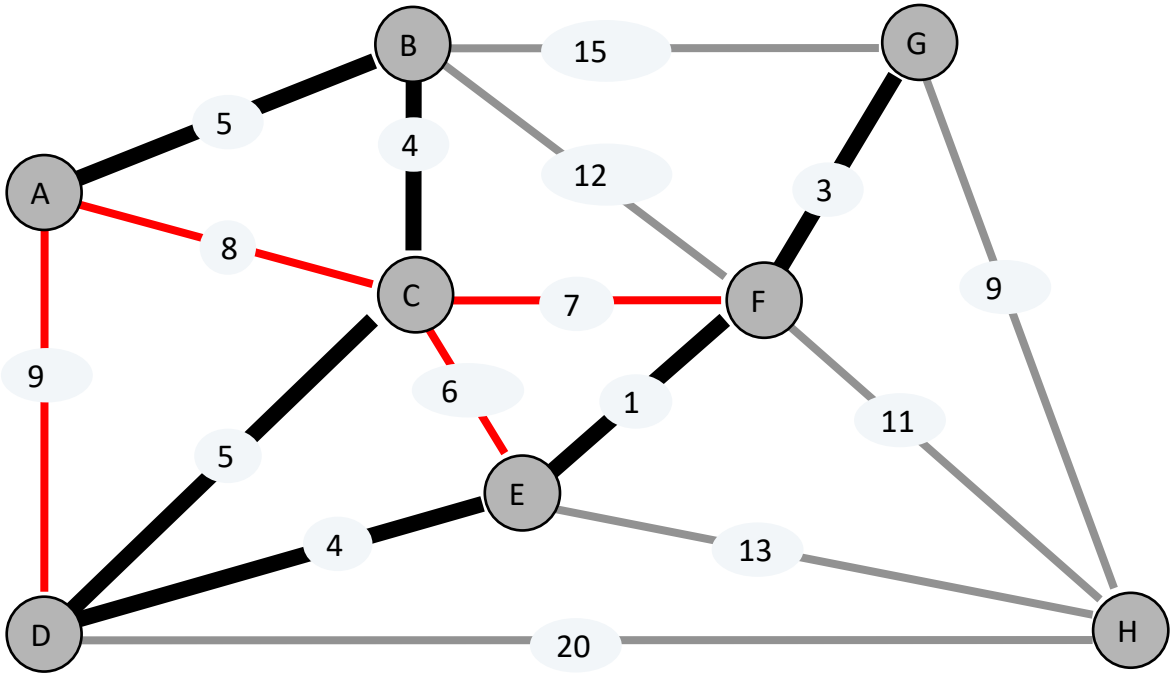
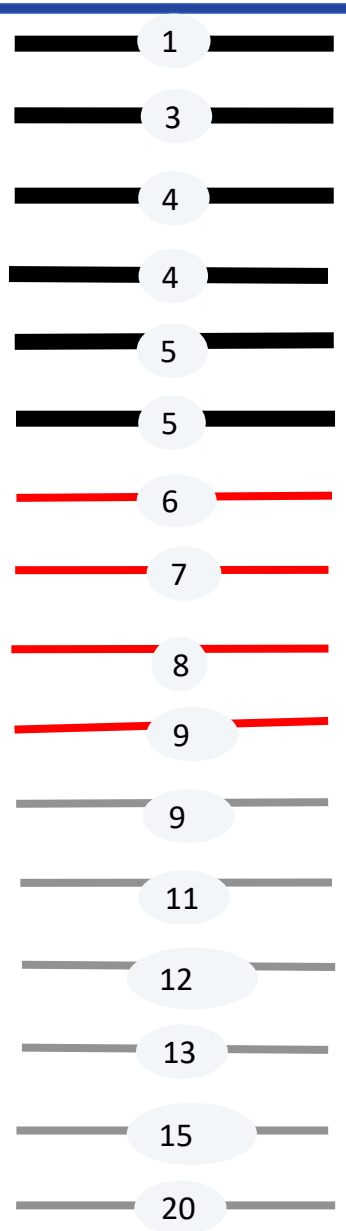
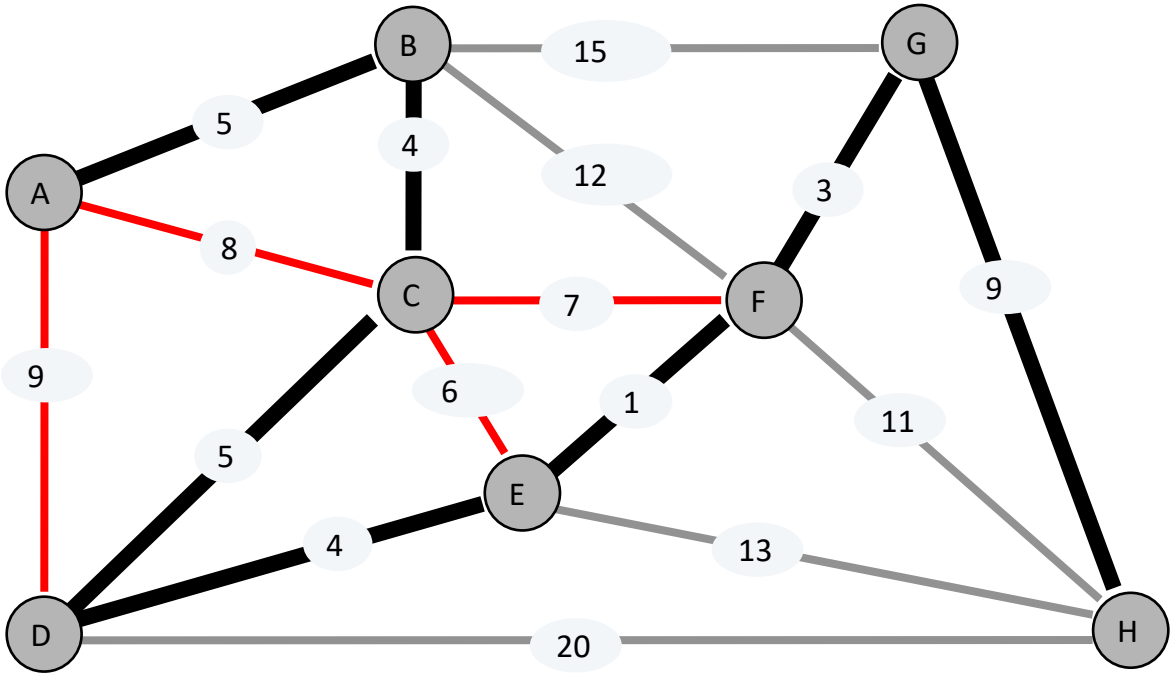
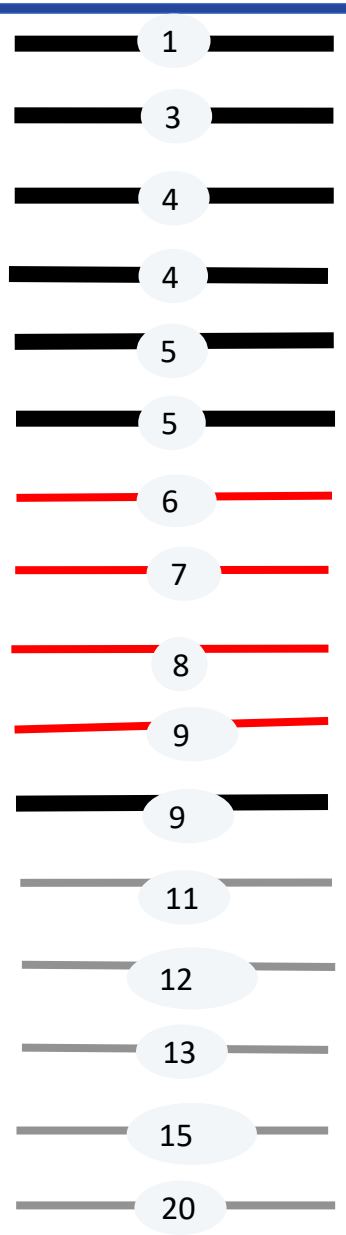




Kruskal's algorithm to find MST





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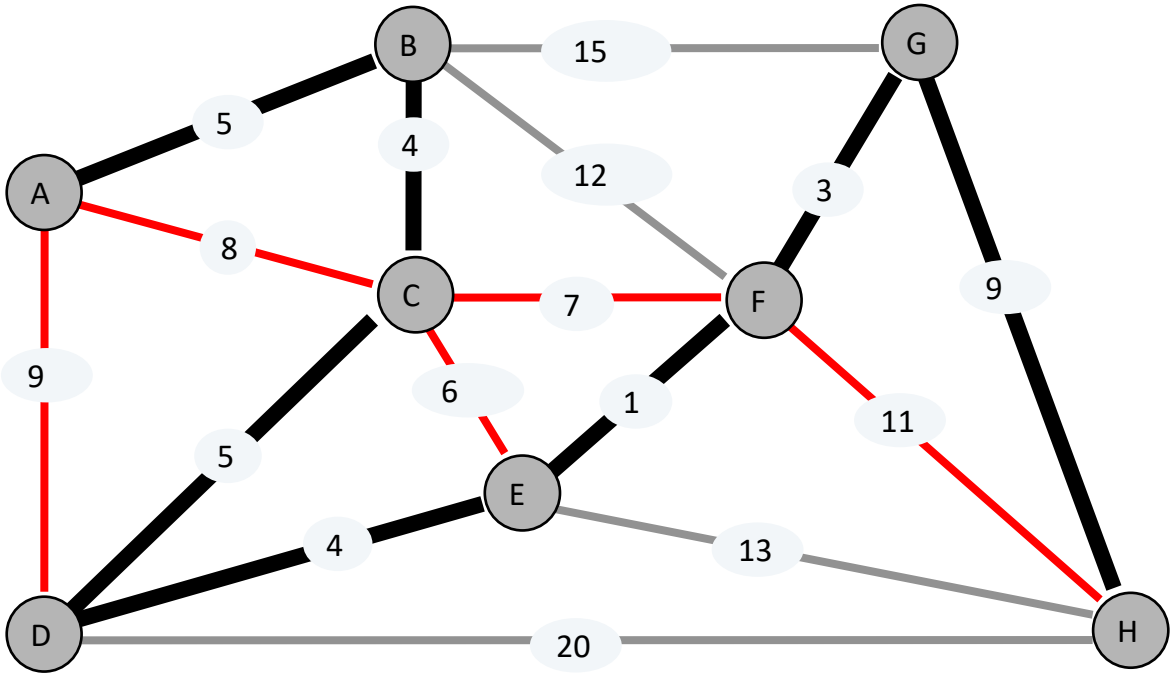

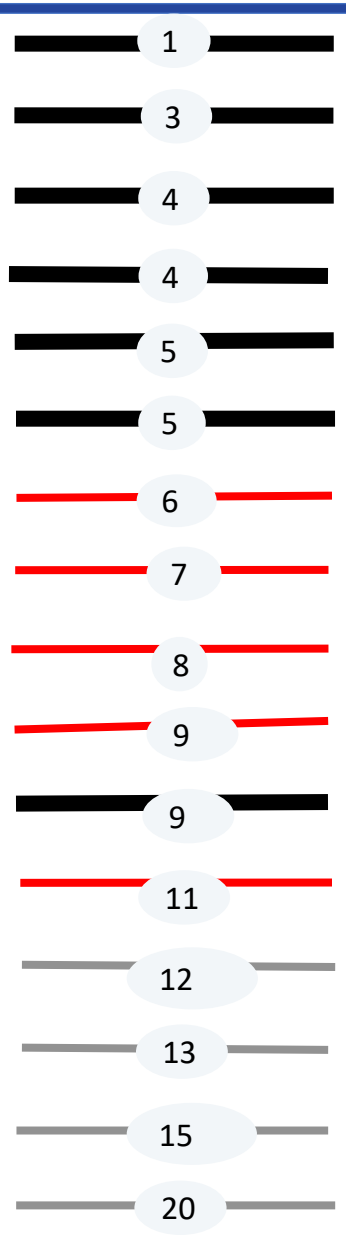

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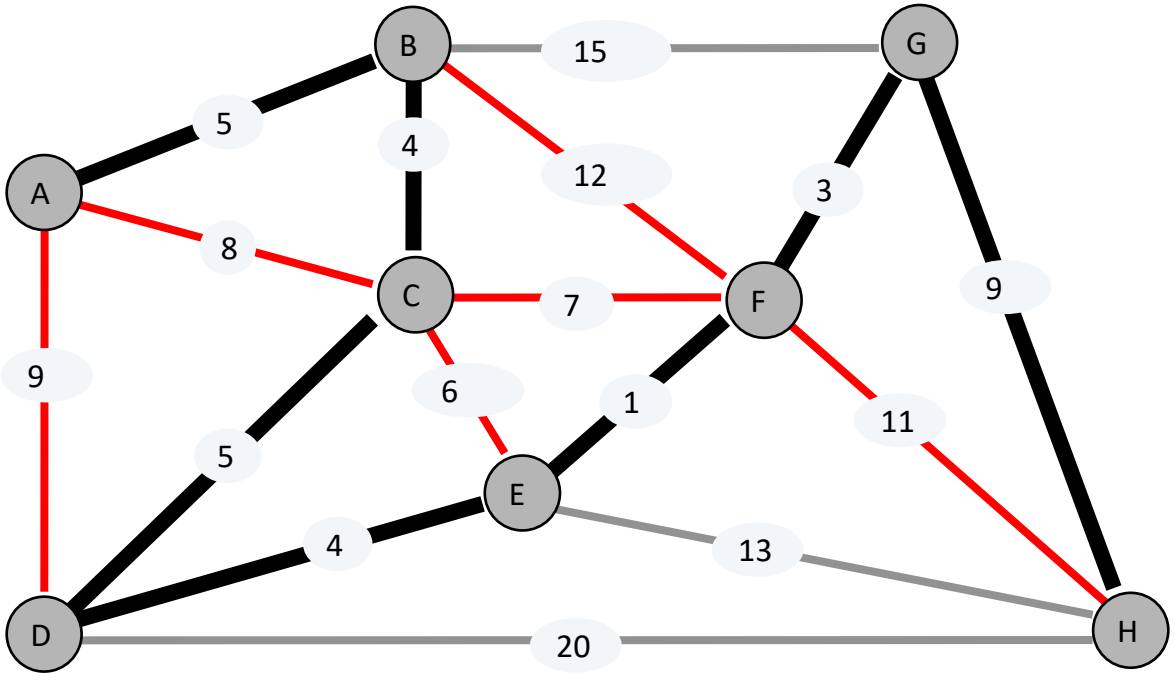

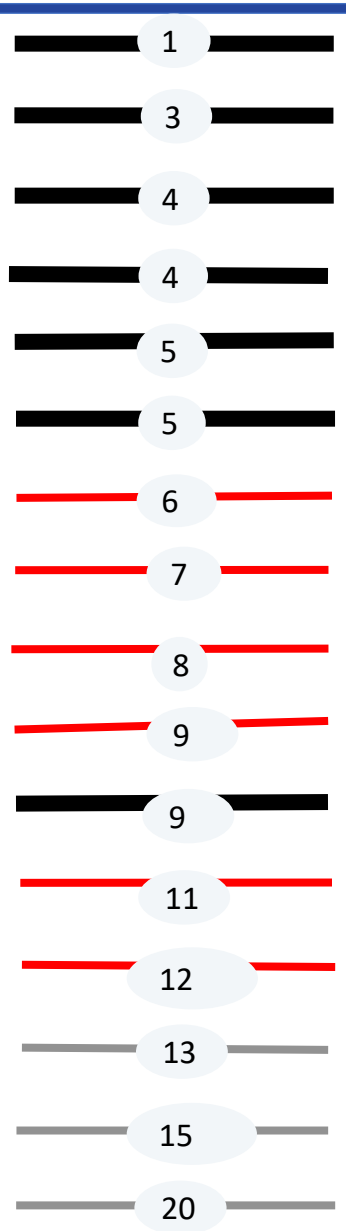

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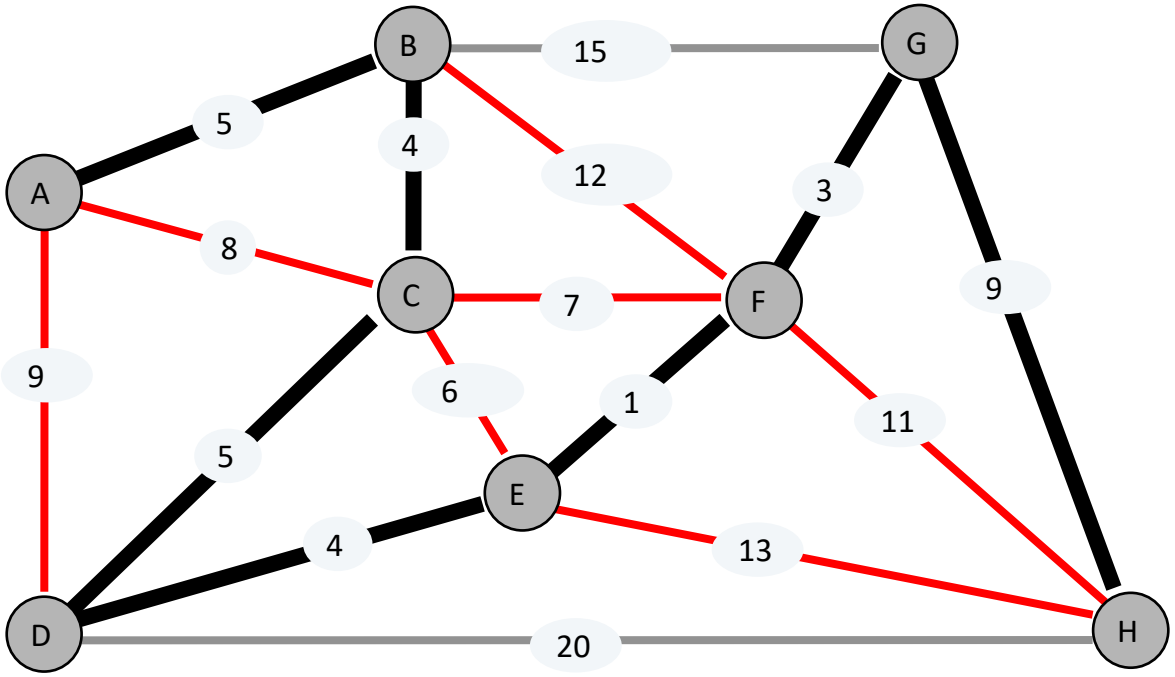

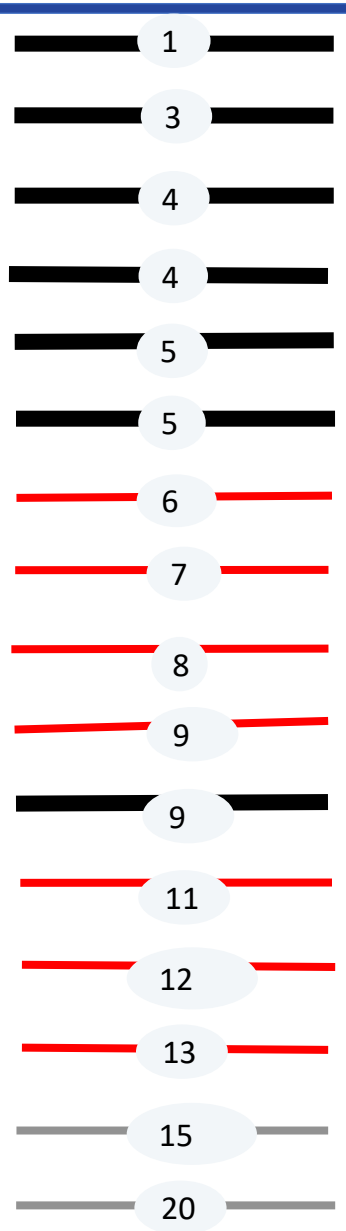

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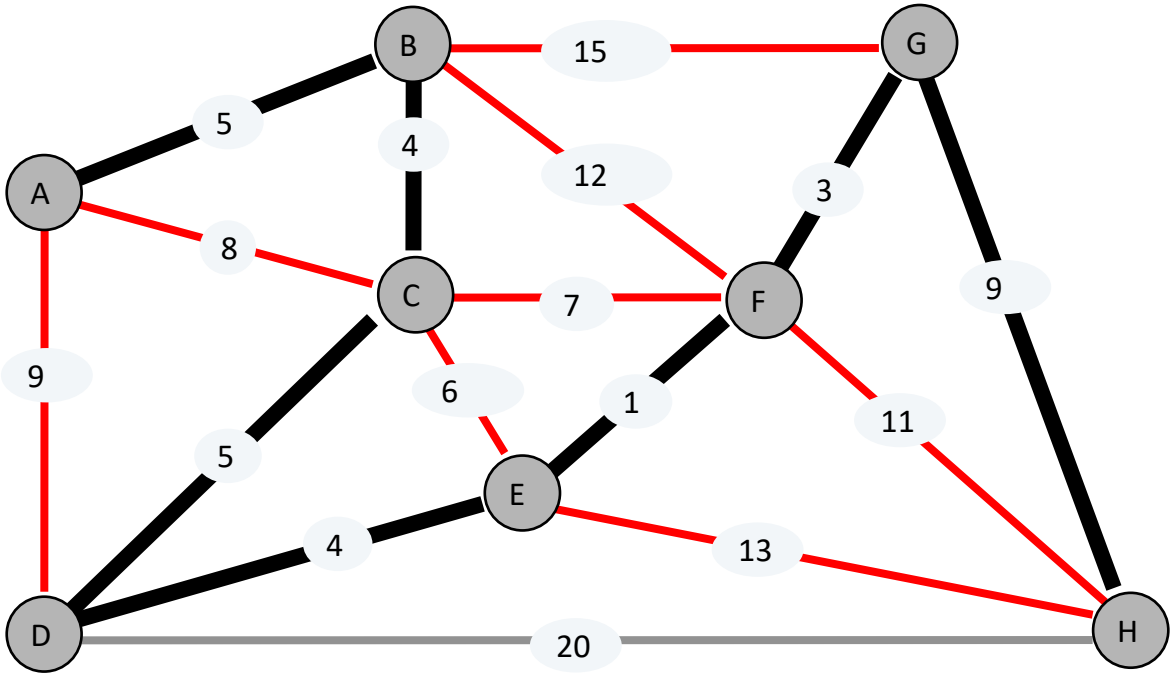

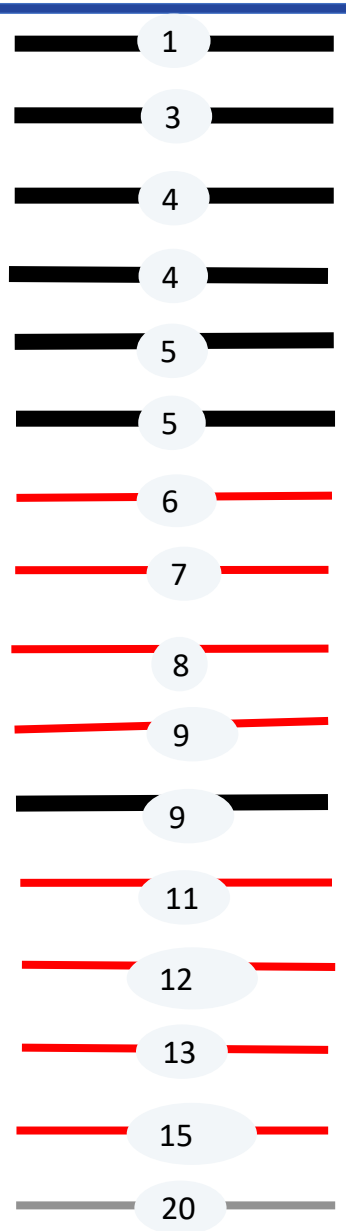

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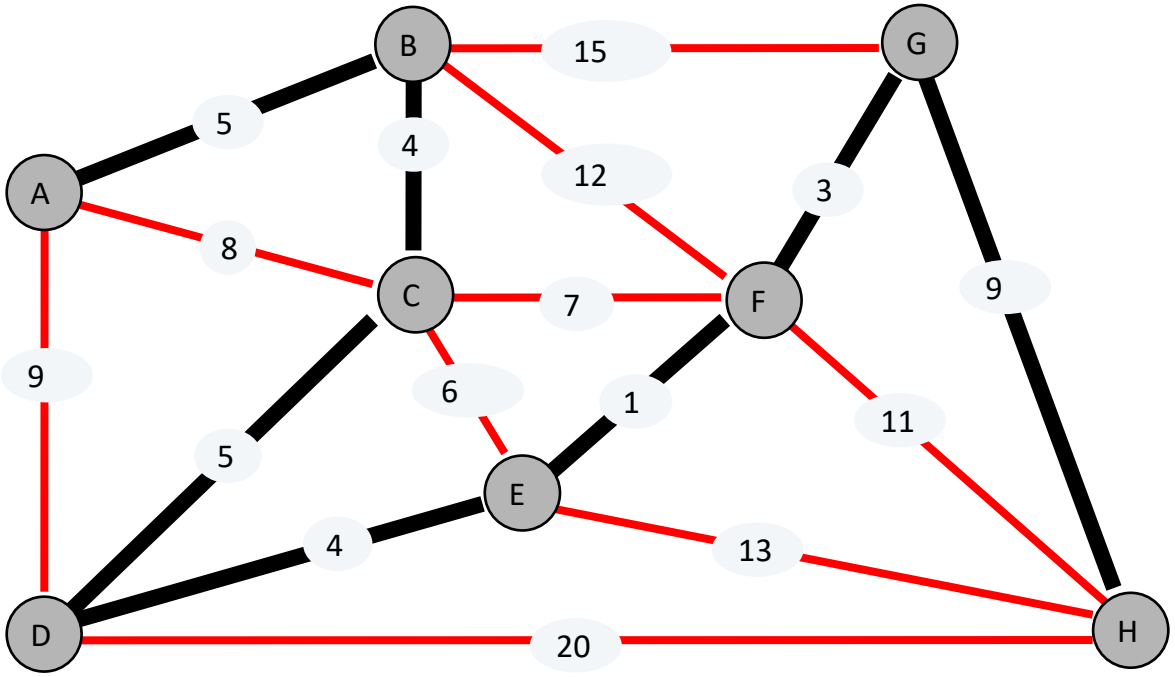
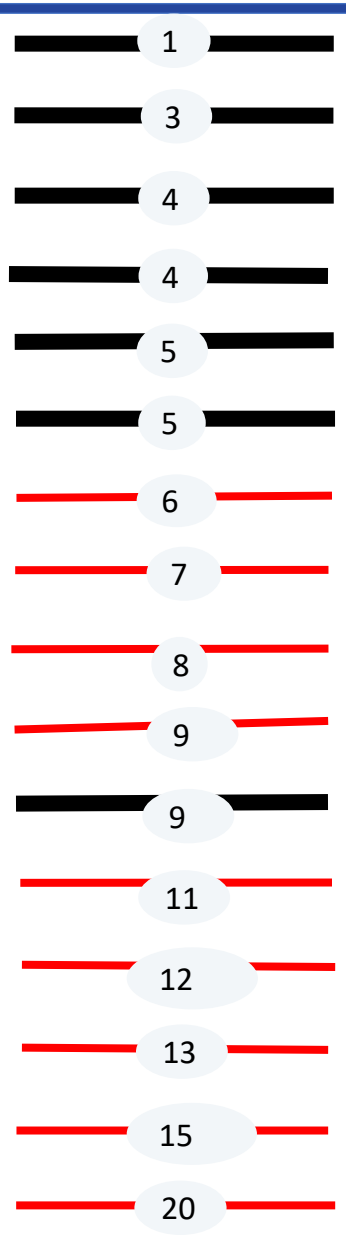
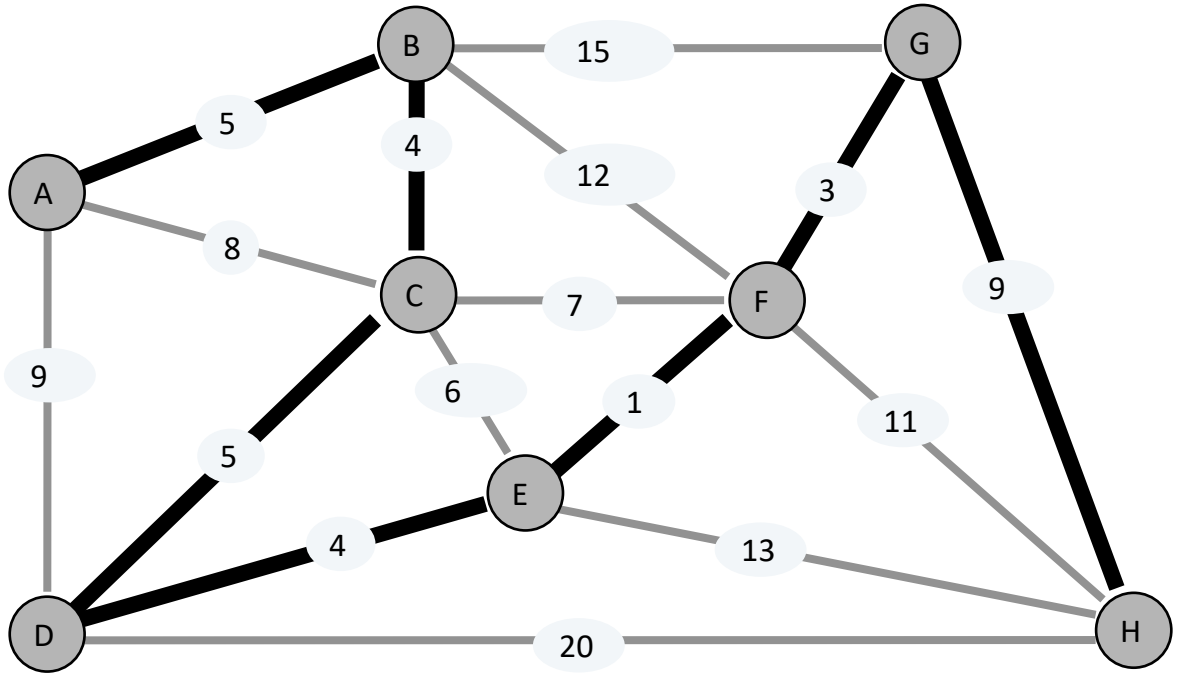
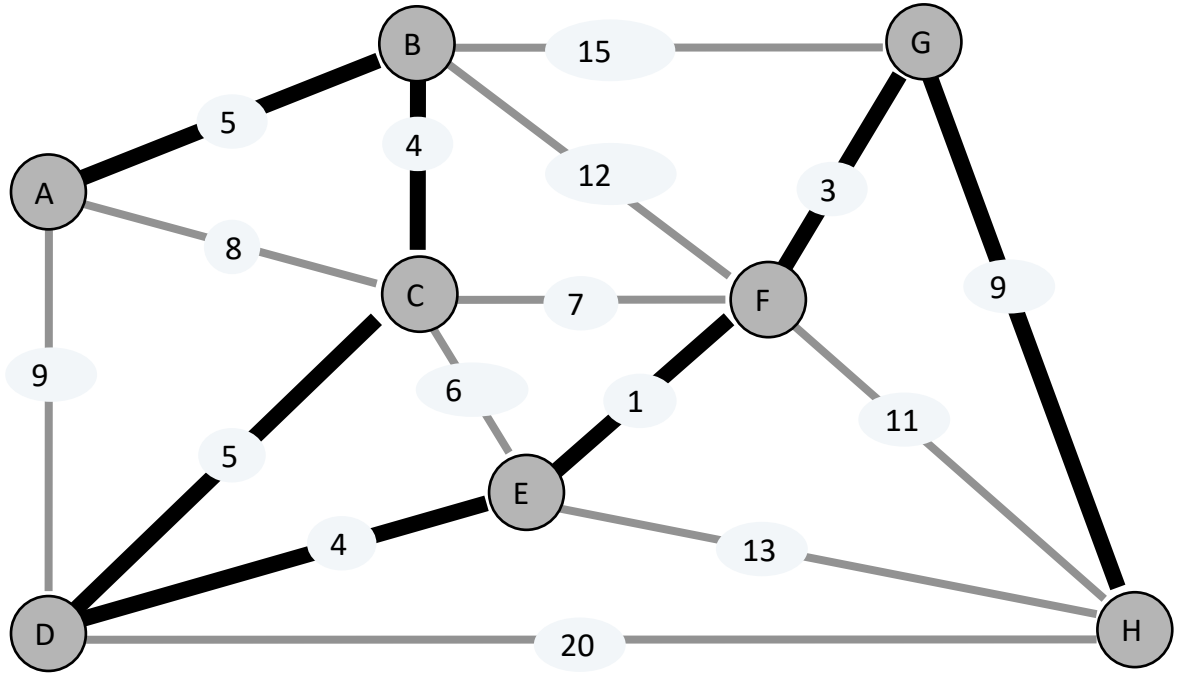




Kruskal's algorithm to find MST

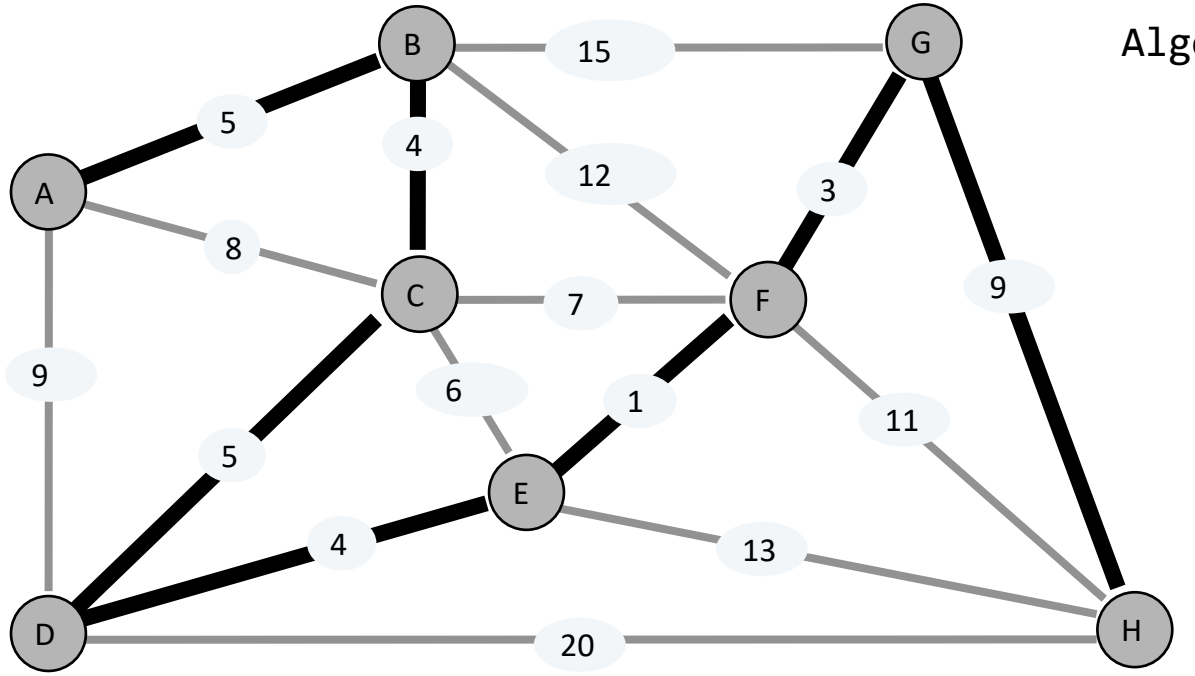


Kruskal's algorithm to find MST



- Sort edges in ascending order of weight.
- Check for each edge
  - **Case 1:** If adding ***e*** to ***MST*** creates a cycle, discard e according to cycle property.
  - **Case 2:** Otherwise, insert ***e = (u, v)*** into ***MST***

Kruskal's algorithm (using BFS/DFS)



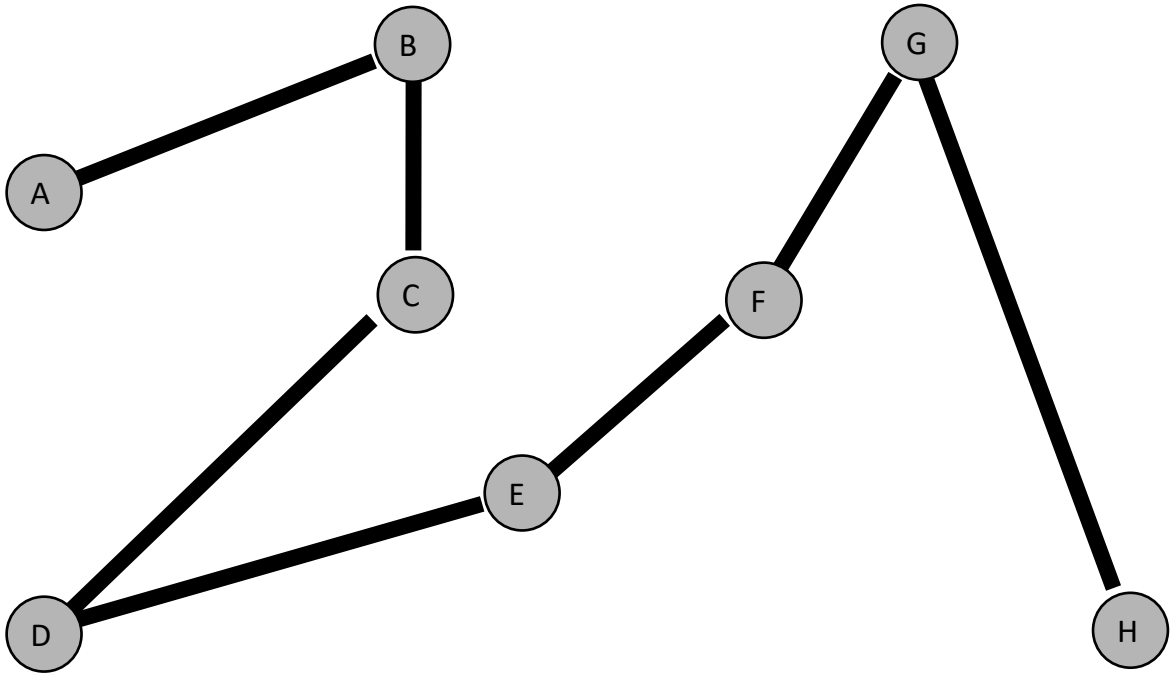

```
Algorithm Kruskal(G):

    MST ← empty set
    Sort edges E by weight in non-decreasing order

    for each edge e = (u, v) in E do
        if not BFS(u, v, MST) then
            Add e to MST
        end if
    end for

    return MST
```

$O(ElogE)$

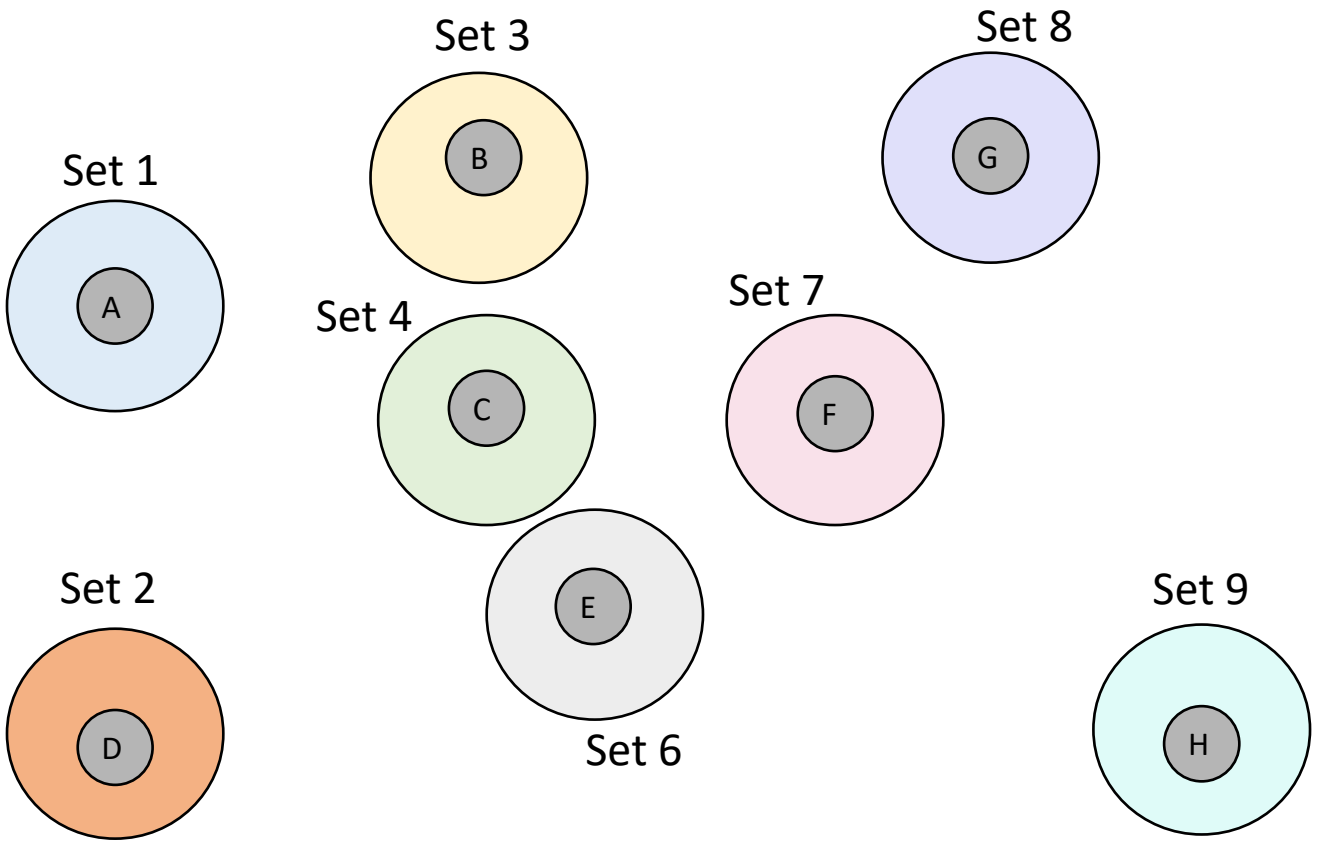
$O(V+E)$

$O(E)$

Kruskal's algorithm (Union-Find Method)

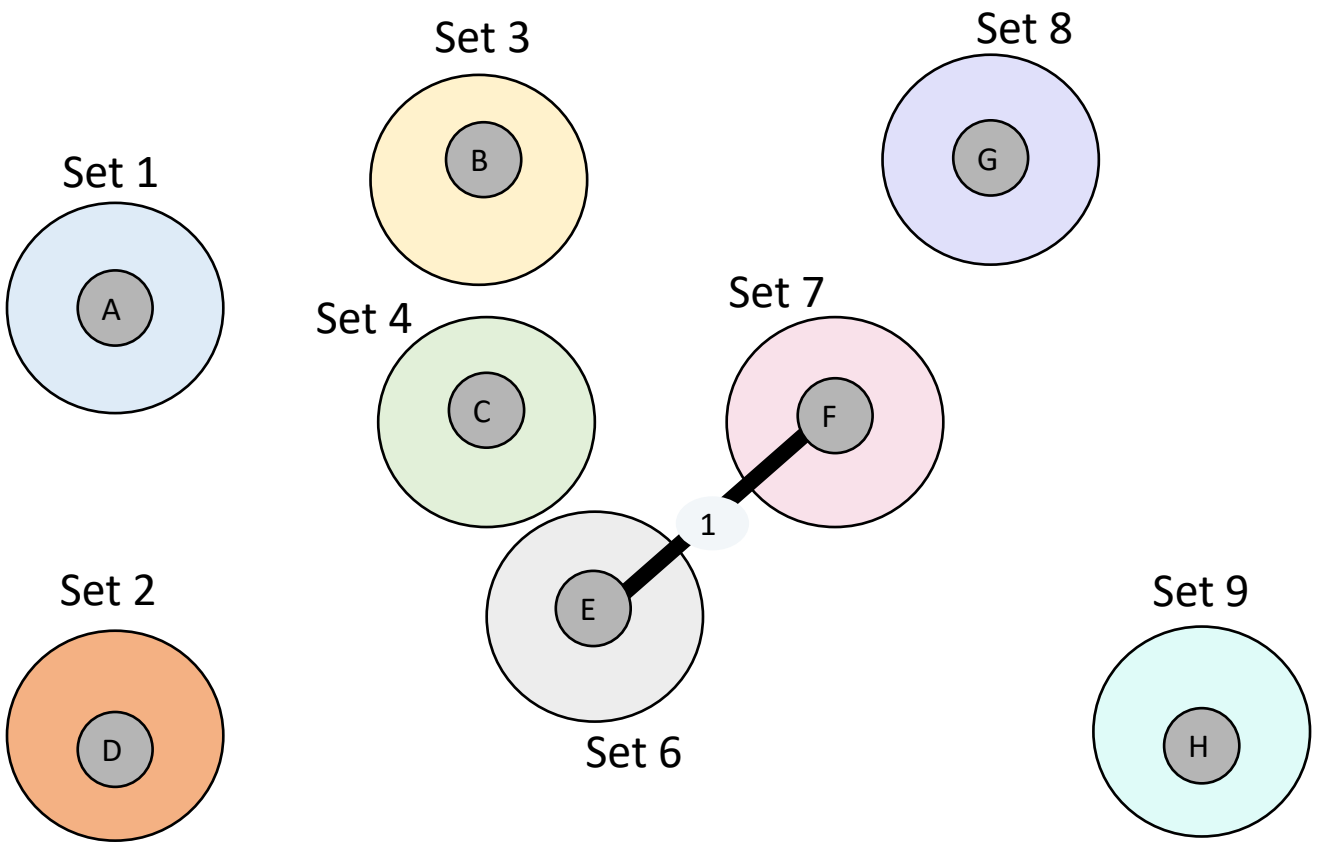


Kruskal's algorithm (Union-Find Method)



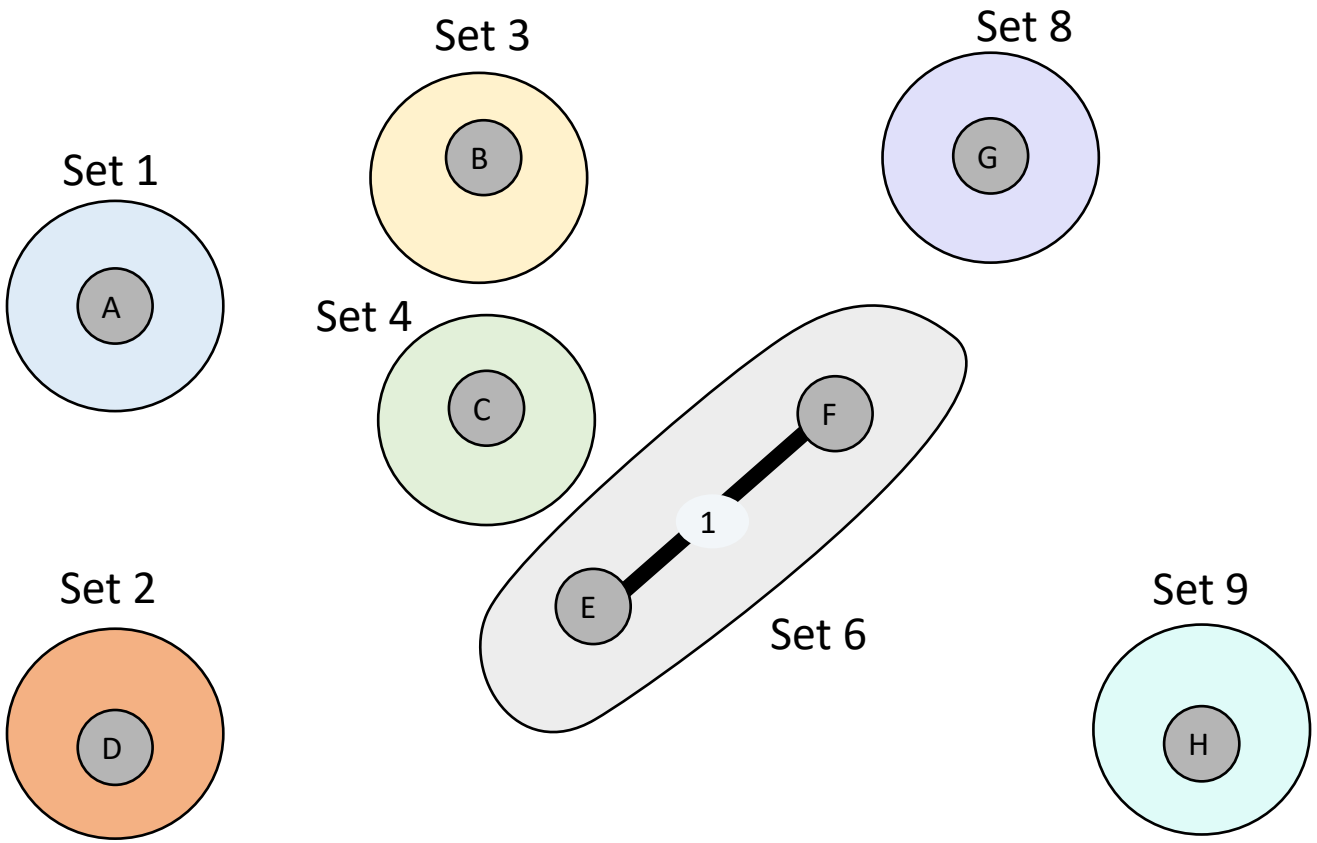
Start with a graph where each vertex is in its own separate set

Kruskal's algorithm (Union-Find Method)



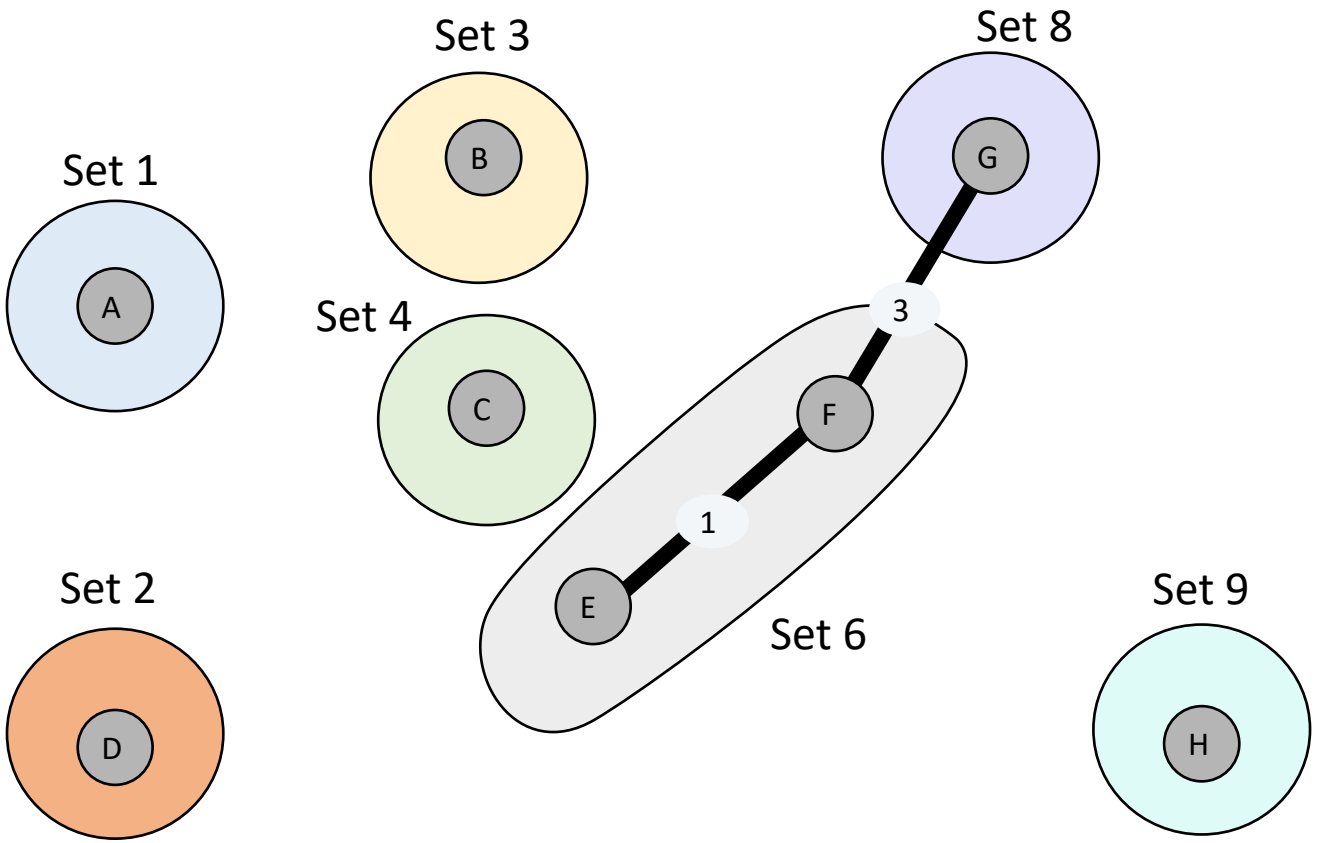
For each edge (which connects two vertices), check if the two vertices belong to the same set or not.

Kruskal's algorithm (Union-Find Method)



If they belong to different sets, it means the edge connects two distinct components of the graph. Perform a union operation to merge the sets of the two vertices

Kruskal's algorithm (Union-Find Method)



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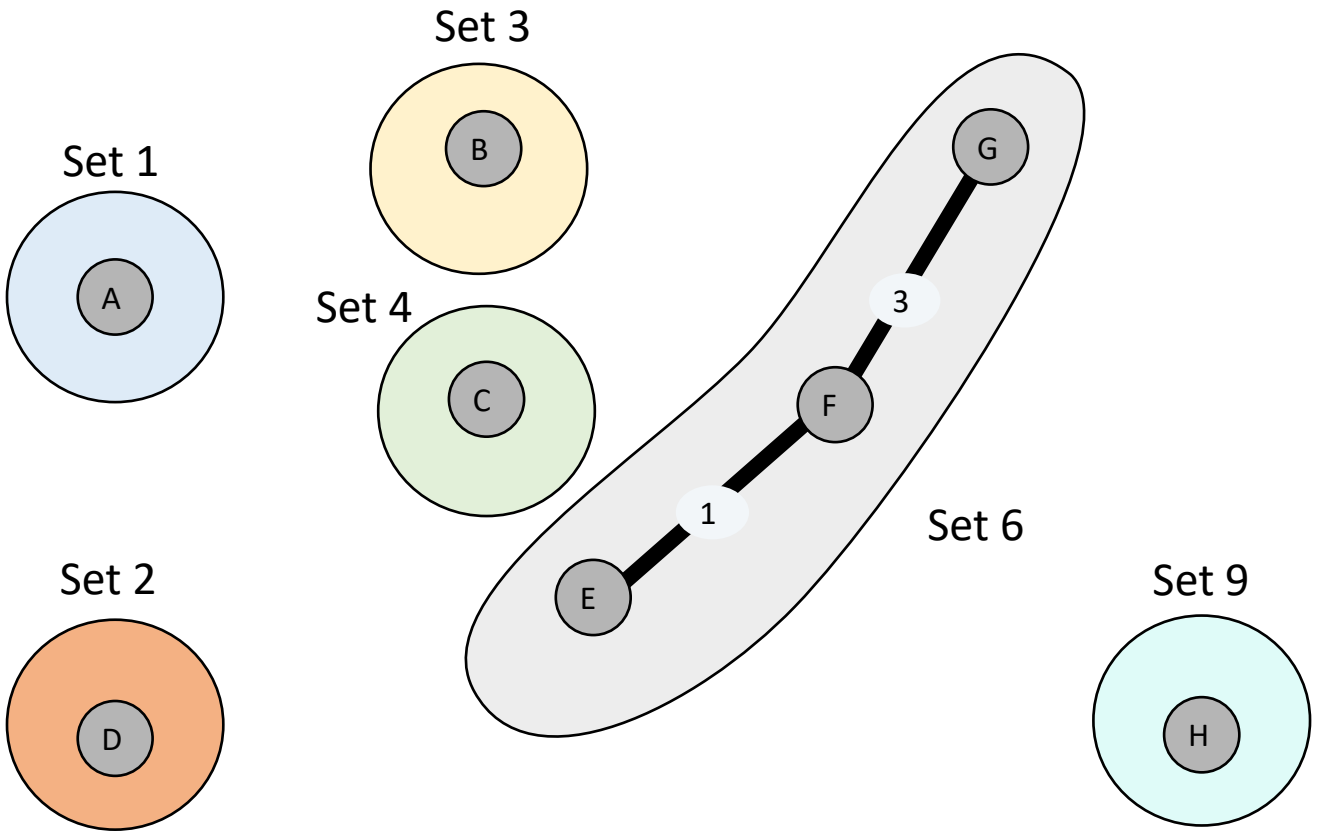
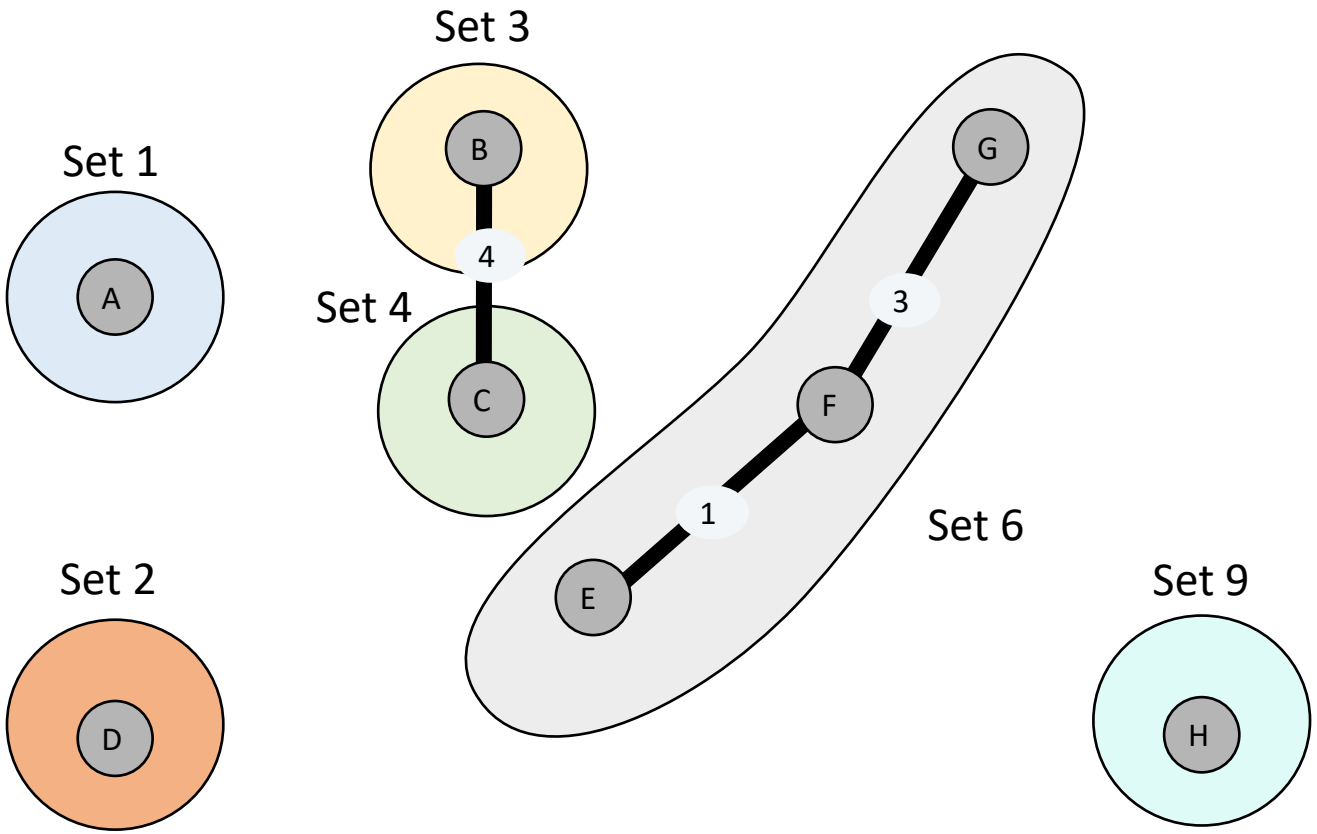
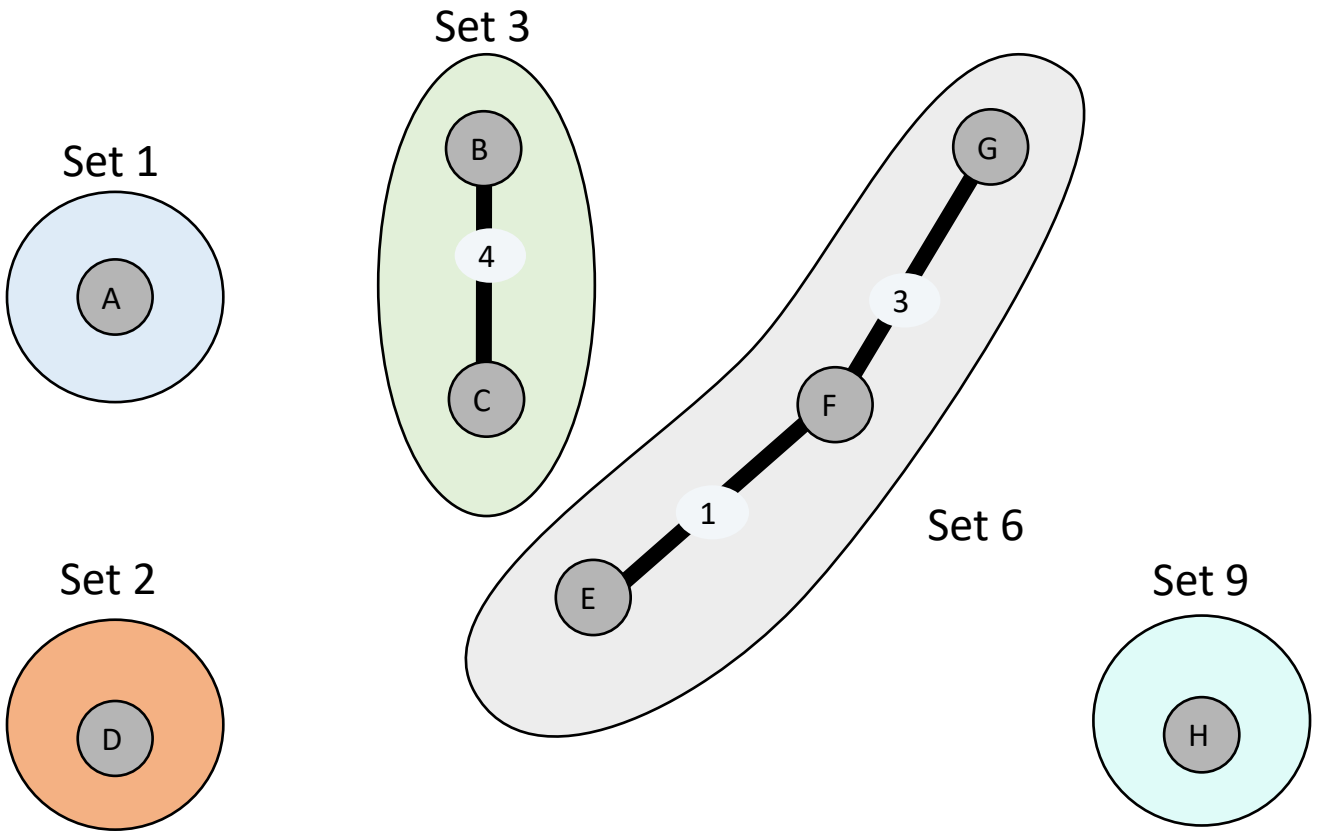


Kruskal's algorithm (Union-Find Method)



Kruskal's algorithm (Union-Find Method)



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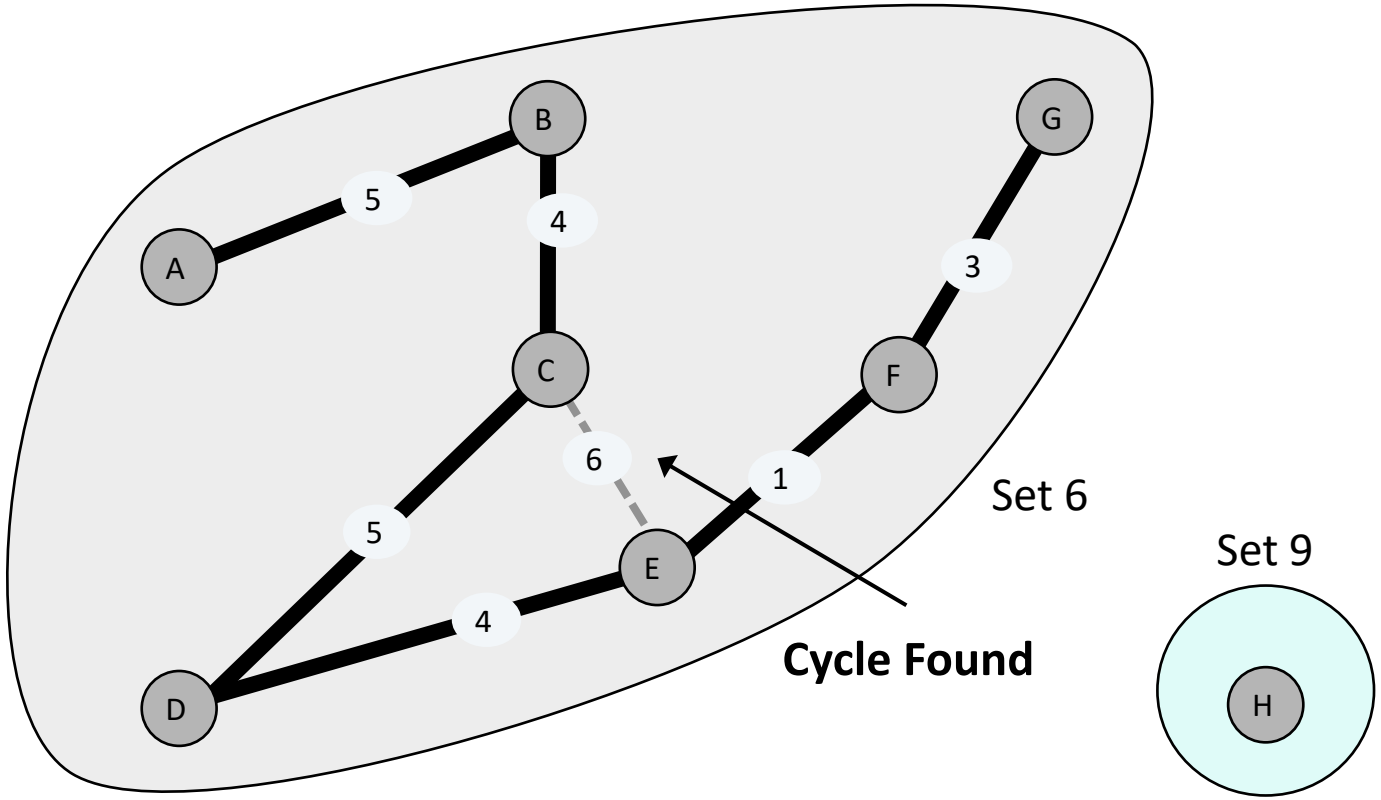


If they belong to the same set, a cycle is detected because adding this edge will create a loop.

Kruskal's algorithm (Union-Find Method)

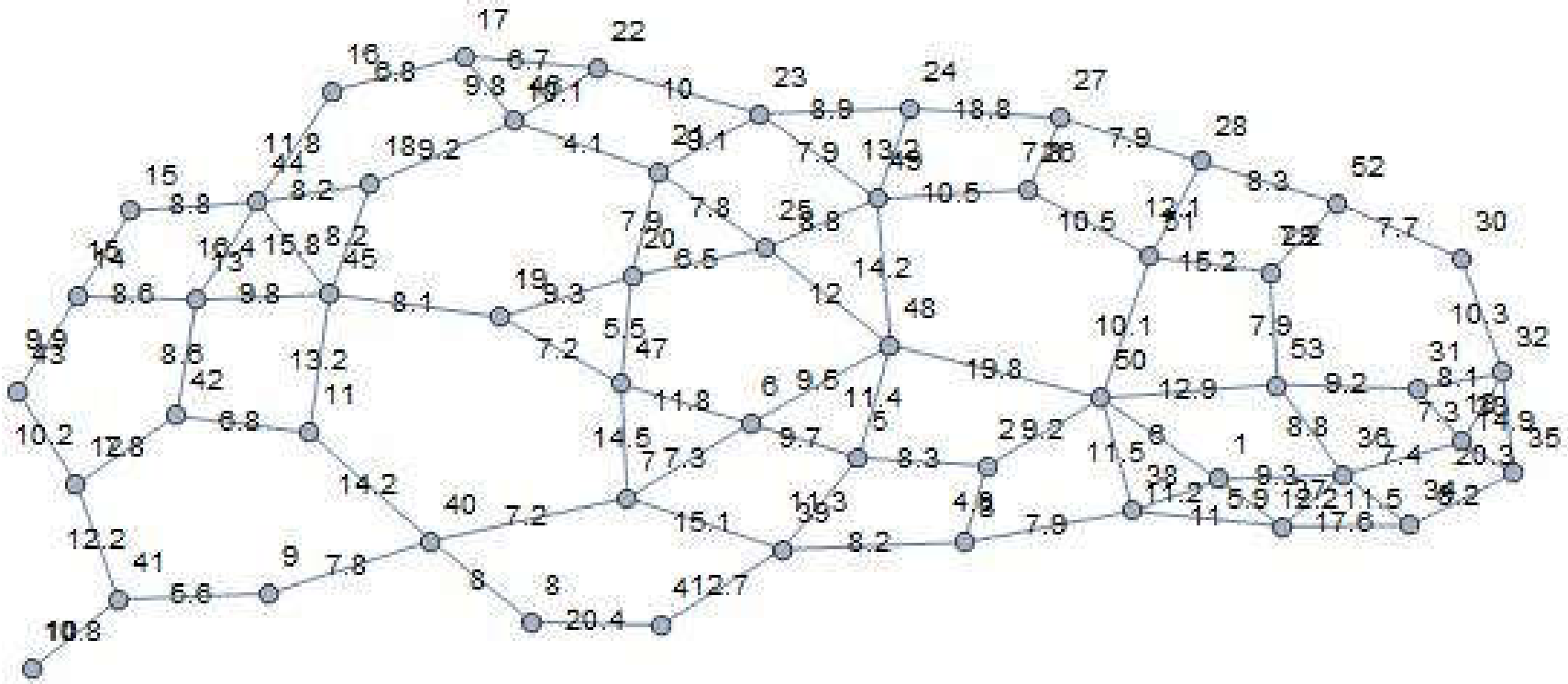
```
Kruskal(G) {
    MST ← empty set
    Sort edges E by weight in non-decreasing order   } O(ElogE)

    foreach (u ∈ V) make a set containing singleton u   } O(V)

    for i = 1 to m
        (u,v) = e_i
        if (u and v are in different sets) {   } O(αV)
            MST ← MST ∪ {e_i}
            merge the sets containing u and v   } O(1)
        }
    return MST
}
```

$O(E)$

Inverse Ackermann function which grows extremely slow. We can consider it as constant

MST: Live Poll 2

Find the minimum spanning tree for the given weighted graph

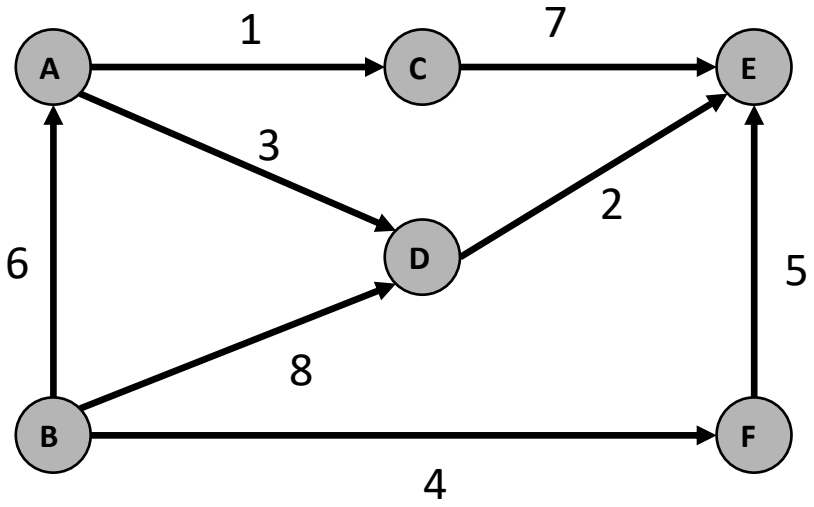


Scan the QR or use link to join

https://forms.office.com/r/MvHNues4DL

MST: Live Poll 2

Find the minimum spanning tree for the given weighted directed graph



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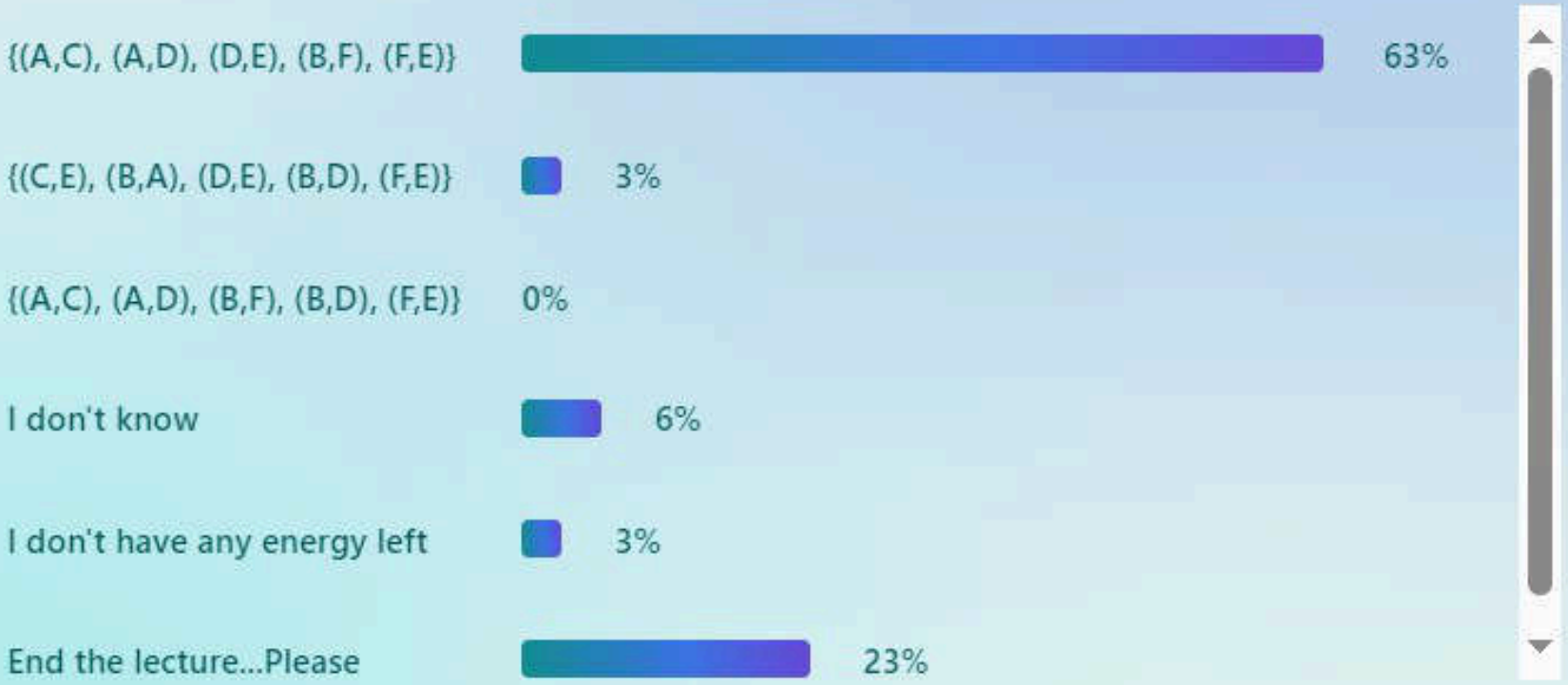
Copy link

30 responses submitted

Find the minimum spanning tree for the given directed graph



Treemap Bar

1 of 1

Thanks a lot

Mathew Perry

"Chandler"

1969 – 2023

If you are taking a Nap, **wake up**.....Lecture Over